Foundation Trilogy

Same-Root, Shared Manifestation, and the One-Step Perfuming Loop

Foundations I, II, and III

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Abstract

This volume collects the three papers of the *Foundation Trilogy* within the *Consciousness Series*. The trilogy treats the source-side or generative layer of the programme: how a source datum is organized, when multiple source chains admit a shared manifest sector, and how a one-step calibration signal may feed back from manifestation to source dynamics.

The trilogy develops a three-step logical chain:

1. **Foundation I: Same-Root Program.** Starting from a Type III_1 source datum, the first paper formulates the conditional source-to-manifest chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

   thereby isolating the theorem-ready interface that later feeds the geometry–modular-flow analysis of the core trilogy.

2. **Foundation II: Shared Manifestation.** The second paper extends the framework to the first multi-source setting. Without assuming any tensor-product factorization, it proves a conditional two-source inclusion theorem for a shared stable hull and shows how a shared admissible manifest sector can sit inside the overlap of two single-source manifest carriers.

3. **Foundation III: One-Step Perfuming Loop.** The third paper introduces a source-level feedback lift driven by manifest-side calibration. Using exact semigroup composition rather than a generator-level perturbation as the defining move, it proves a conditional one-step closed-loop foothold and a controlled deformation of a supplied manifest-side comparison functional. It does not import a nonzero manifest deficit floor as an established theorem.

Taken together, the three Foundation papers supply the generative-side scaffolding of the programme: a conditional same-root bridge, a conditional shared-sector foothold, and a conditional one-step return arrow from manifest calibration to source dynamics.

Keywords: open quantum systems, quantum dynamical semigroups, von Neumann algebras, Type III_1 factors, modular theory, same-root structure, shared manifestation, source-level feedback

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Foundation I: From Source to Manifestation
A Conditional Same-Root Program for Geometry and Modular Flow

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Version 2

Abstract

Paper I of the Consciousness Trilogy [1] begins from an admissible accessible algebra $(\mathcal{A}_c, \omega)$ and derives the inseparability of geometry and modular flow. The deeper source-side question is how one reaches such an object from the pre-observer substrate developed across Genesis, especially the HAFF and T-DOME papers. The present paper, Foundation I of the Consciousness Series, does not claim a completed unified-source theorem. Instead, it formulates an honest source-side foundation program and pushes its currently closed parts into paper form.

We first state the correct conditional target: if a persistent Nakajima–Zwanzig projection exists in the source datum, and if the foreground-to-algebra transfer can be carried out, then the chain

$$S_{\text{source}} \to \mathcal{K}_* \to \mathfrak{F} \to (\mathcal{A}_c, \omega) \to (\text{Geom}, \text{Mod})$$

is forced. Within that program, we prove the packaging result that budget violation in the persistent record process forces symmetry-broken foreground compression, thereby rewriting the T-DOME I–II chain as a clean Foundation I proposition. We then prove a first restricted PS2 closure in a product-classical, decoherence-dominated pointer regime: pointer coarse-graining yields fragment-local omission losses, semigroup decay yields a uniformly bounded coherence remainder, and a weight gap larger than the resulting remainder forces exact top-k^* pointer selection. In a first finite-dimensional faithful pointer-support regime, the required coherence constants become explicit in Hilbert–Schmidt norm. We then define a pointer-aligned realization of the foreground frame inside the ambient algebra, construct the associated pointer-generated von Neumann algebra, and replace the earlier open-ended transfer candidate by a minimal modular-decoherence stable hull $\Phi^{\text{st}}_{\mathfrak{F}}(\mathfrak{F})$. This stable-hull transfer is well-defined, yields a von Neumann subalgebra, and carries a faithful restricted state.

Finally, under the exact fixed-point form of the HAFF accessibility condition, we prove the inclusion-level comparison

$$\Phi^{\text{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\text{HAFF}},$$

while leaving equality as an open strong-form problem. The upshot is deliberately calibrated: Genesis already supplies substantial antecedent support for Foundation I, but the

completed deductive source-side foundation still depends on two genuinely open interfaces, namely persistent projection selection and the model-independent derivation of PS2 beyond the restricted product-classical pointer regime treated here.

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1 Introduction

Paper I of the Consciousness Trilogy [1] starts at the manifest layer: one assumes an admissible accessible algebra $(\mathcal{A}_c, \omega)$, proves that geometry and modular flow are not independently deformable, and then studies the consequences of that inseparability. That was the correct place for the first trilogy paper to begin, but it leaves a source-foundation question unresolved. What, in the broader Genesis programme, plays the role of the source-side antecedent to $(\mathcal{A}_c, \omega)$?

The present paper, Foundation I of the Consciousness Series, addresses exactly that question. It does not claim that Genesis already contains a completed theorem deriving the trilogy from a single source object. What it does claim is narrower and more precise: the relevant source-side foundation can now be decomposed cleanly, its nearly closed parts can be written as formal propositions, and its actual bottlenecks can be localized without conflating vision and theorem.

The philosophical motivation is correspondingly limited. Paper I showed that geometry and modular flow are not independently deformable once one is already at the manifest pair $(\mathcal{A}_c, \omega)$. Foundation I asks the prior structural question: why should those two readings belong together at all? In the present paper, the same-root idea means only that one seeks a source-side chain from which geometric and modular aspects later arise together. This paper treats that mathematical bridge from source data to accessible algebra; it does not attempt to settle consciousness ontology or the broader philosophy of mind.

The correct three-layer structure is

$$S_{\text{source}} \longrightarrow (\mathcal{A}_c, \omega) \longrightarrow (\text{Geom}, \text{Mod}),$$

with an important internal refinement:

$$S_{\text{source}} \longrightarrow \mathcal{O}_A \longrightarrow \mathcal{O}_B \longrightarrow (\mathcal{A}_c, \omega).$$

Here S_{source} is the pre-observer source datum, $\mathcal{O}_A$ is a persistent projected record process with non-Markovian memory, and $\mathcal{O}_B$ is the symmetry-broken foreground frame produced when full-rank persistence exceeds the available processing budget.

Four claims structure the paper.

1. We state the honest *conditional* Foundation I target. The two genuine open points are persistent projection selection and the model-independent foreground-alignment step PS2 that makes the transfer construction canonical; the existence of an exact fixed-point pointer sector is treated only as an exact-regime PS1 idealization.

2. We isolate a finite-coordinate coding regime in which a binding rank budget and an irreducible symmetry action yield a symmetry-breaking foreground choice. This is a conditional model result, not an import of a universal T-DOME necessity theorem.

3. We prove a first restricted PS2 closure package: in a product-classical, decoherence-dominated pointer regime, pointer coarse-graining yields fragment-local omission losses, semigroup decay yields a uniformly bounded coherence remainder, and a weight gap larger than the resulting remainder forces exact top-k^* pointer selection.

4. We give the paper's main new operator-algebraic construction: a stable-hull realization of the foreground sector. Concretely, we construct a minimal modular-decoherence stable hull, prove that it is a von Neumann subalgebra with faithful restricted state, and prove the first comparison theorem $\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\text{fix}}$ under an explicit fixed-set algebra-closure condition.

This is enough to fix the current calibration. Genesis provides substantial antecedent support for Foundation I. It does not yet provide a finished deductive source-side foundation. No equality between the foreground-generated exact fixed sector, probe-accessible effects, and the environmental record algebra is claimed here.

The paper is organized as follows. Section 2 defines the source datum and the stage outputs. Section 3 states the target conditional theorem and isolates the source-selection problem. Section 4 proves the conditional finite-coordinate compression proposition. Section 5 defines pointer alignment, foreground realization, and the PS2 target. Section 6 proves a first restricted PS2 closure package in the product-classical, decoherence-dominated pointer regime, and records a bookkeeping lemma fixing the diagonal/off-sector split needed for the later model-independent PS2 programme. Section 7 proves the first inclusion-level comparison with the typed exact fixed sector and states a partial Lemma C. Section 8 records the remaining open problems. Appendix A summarizes the external Genesis inputs used in the paper in self-contained mathematical form, and Appendix B collects the named conditional interfaces that remain open.

2 Source Datum and Object Ledger

Definition 2.1 (Source datum)**.** A *source datum* is a quadruple

$$S_{\text{source}} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

where:

1. $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a Type III_1 von Neumann algebra in standard form, with Ω cyclic and separating for $\mathcal{A}_U$;

2. $\{\mathcal{E}_t\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$.

Remark 2.2 (Heisenberg-picture continuity convention). The semigroup $\{\mathcal{E}_t\}$ acts on observables, so the continuity assumption in Definition 2 is pointwise σ-weak continuity on $\mathcal{A}_U$, not norm continuity on the ambient algebra. This is the natural continuity regime for the Heisenberg-picture QDS language used throughout the paper.

The ambient standard-form side is the HAFF input [3]; the projected open-system side is expressed in Nakajima–Zwanzig language [8, 11, 7].

Definition 2.3 (Stage outputs)**.** Given a source datum S_{source}, we use the following ledger.

1. **Stage A output**
$$\mathcal{O}_A := (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s)),$$
where $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$ is a Nakajima–Zwanzig projection, $\omega_{\mathcal{P},t} := \mathcal{P}_*(\omega_t)$ is the projected normal state on the predual, and $\mathcal{K}_*(t,s)$ is the dual memory kernel.

2. **Stage B output**
$$\mathcal{O}_B := (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s), \mathfrak{F}, \pi_{\text{ego}}),$$
where $\mathfrak{F}$ is a k^*-dimensional foreground frame and π_{ego} is the ego prior carried forward from the broken phase of T-DOME II and into the self-calibration analysis of T-DOME III [5, 6].

3. **Stage C output**

$$\mathcal{O}_C := (\mathcal{A}_c, \omega), \qquad \omega = \langle \Omega, \cdot\, \Omega \rangle|_{\mathcal{A}_c}.$$

Remark 2.4 (Stage D). Once an admissible $(\mathcal{A}_c, \omega)$ is available, the Paper I chain from accessible algebra to jointly constrained geometry and modular flow is already in place [1]. The present note is therefore concerned only with Stages A–C.

3 Target Conditional Theorem and the Source-Selection Problem

The honest Foundation I target is conditional and should be stated that way from the start.

Target conditional theorem. Let

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t \geq 0})$$

be a source datum. Assume:

1. there exists a persistent Nakajima–Zwanzig projection $\mathcal{P}$ whose projected predual dynamics carries a non-Markovian kernel $\mathcal{K}_*(t, s)$;
2. a specified finite-coordinate coding model has a binding rank budget and satisfies the optimization and group-action hypotheses of Proposition 4.1, producing a compressed foreground frame $\mathfrak{F}$;
3. there exists a foreground embedding

$$\iota_{\mathfrak{F}} : V_{\text{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\text{sa}}$$

   from which the transfer construction can be carried out and compared, at the inclusion level, with the separately typed exact fixed sector;
4. the resulting $(\mathcal{A}_c, \omega)$ satisfies the admissibility hypotheses needed by Paper I.

Then the chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega) \to (\text{Geom}, \text{Mod})$$

is defined conditional on assumptions (1)–(4). No arrow in this display is asserted to follow from the bare source datum alone.

Remark 3.1 (What is actually open). The conditional theorem has several independent burdens. They include existence and selection of a persistent projection $\mathcal{P}$, the model-specific coding and frame-selection package in Proposition 4.1, and the transfer from the foreground frame to the operator-algebraic stable hull. Paper I shell generation is a further interface input rather than a Foundation I consequence. These burdens must not be compressed into a single claimed necessity chain.

Lemma 3.2 (Conditional source-to-kernel interface)**.** *Let S_{source} be a source datum. Suppose there exists a Nakajima–Zwanzig projection $\mathcal{P}$ such that the projected predual dynamics admits a non-Markovian kernel $\mathcal{K}_*(t, s)$ and yields a persistent record process. Then*

$$\mathcal{O}_A = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t, s))$$

satisfies the input conditions of Stage B.

Proof. This is the conditional form of the source-side interface. The assumed projection already provides the Stage A output. Within the T-DOME I modeling framework [4], the persistence assumption is analyzed in a non-Markovian Nakajima–Zwanzig regime; the present paper imports that framing rather than claiming it as a general open-quantum-systems theorem. The point is not to solve the selection problem constructively, but to separate that problem from the downstream thermodynamic and algebraic consequences. □

Remark 3.3 (Constructive version remains open). Lemma 3.2 is intentionally conditional. The missing constructive question is why a given source datum selects a persistent projection at all. Nothing in the present note should be read as claiming that this observer-emergence problem has been solved.

4 Conditional Foreground Compression

We now isolate a finite-coordinate model in which the foreground choice is a valid consequence of explicit coding assumptions. The revised T-DOME results do not establish that memory is necessary for persistence, that every processing budget is saturated, or that every compression problem must break a symmetry. The proposition below therefore states the extra assumptions directly.

Proposition 4.1 (Conditional finite-coordinate foreground compression)**.** *Let*

$$\mathcal{O}_A = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s))$$

be a persistent record process arising from a source datum S_{source}*, carrying a specified real coordinate space* V *of dimension* D*. Let* $h_\mu > 0$ *be the stipulated processing rate per retained coordinate. Let admissible encoders be orthogonal projections* Q *on* V*, with processing cost* $h_\mu \operatorname{rank} Q$*, and assume:*

1. *the full-rank encoder is infeasible,*

$$h_\mu D > \dot{\mathcal{C}}_{\text{budget}};$$

2. *the survival-distortion functional attains its minimum on the feasible projections and is strictly improved whenever a feasible projection can be enlarged by one retained coordinate;*

3. *a group* G *acts irreducibly and orthogonally on* V*, the survival-distortion functional is* G*-invariant, and* $0 < \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor < D$*.*

Then every minimizing encoder has rank

$$k^* = \left\lfloor \frac{\dot{\mathcal{C}}_{\text{budget}}}{h_\mu} \right\rfloor < D.$$

Its range defines a k^**-component foreground frame* $\mathfrak{F}$*. No such proper nonzero projection is* G*-invariant, so choosing a minimizer selects a frame and breaks the stipulated* G*-symmetry.*

Proof. Feasibility gives $\operatorname{rank} Q \leq \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor$. By strict improvement under every feasible rank-one enlargement, no minimizer can have smaller rank. Hence every minimizer has rank

$$k^* = \Big\lfloor \frac{\dot{\mathcal{C}}_{\text{budget}}}{h_\mu} \Big\rfloor.$$

The range of a minimizing projection supplies the foreground subspace. If that projection commuted with the irreducible G-representation, its range and kernel would be invariant subspaces. Irreducibility would then force the projection to be either 0 or 1, contrary to $0 < k^* < D$. Thus each minimizing choice breaks the stipulated symmetry. This proof uses only the stated coding model; T-DOME I–II motivate the variables but supply no universal necessity premise here [4, 5]. □

Remark 4.2 (Status of Proposition 4.1). The proposition is an FD/I model result. Its rank charge, strict improvement, and irreducible group action are explicit inputs. In particular, it does not say that persistence requires memory, that a general optimum saturates a budget, or that SSB is unavoidable outside this encoder class.

5 Foreground Realization inside the Ambient Algebra

Lemma C requires a genuinely new object: a way to realize the foreground frame inside the ambient algebra.

Definition 5.1 (Exact and finite-window pointer alignment)**.** A self-adjoint observable $O \in \mathcal{A}_U^{\text{sa}}$ is called *pointer-aligned* with respect to the decoherence semigroup $\{\mathcal{E}_t\}_{t\geq 0}$ if it is exactly fixed:

$$\mathcal{E}_t(O) = O \qquad \forall\, t \geq 0.$$

For an explicitly approximate finite-window statement, define

$$d_\tau(O) := \sup_{0\leq t\leq \tau} \|\mathcal{E}_t(O) - O\|.$$

We call O (ε, τ)-pointer-aligned when $d_\tau(O) \leq \varepsilon$.

Remark 5.2 (Why the former asymptotic definition was not weaker). For a bounded semigroup, the former condition $\lim_{t\to\infty} \|\mathcal{E}_t(O) - O\| = 0$ already implies exact fixation. Indeed, for fixed $s \geq 0$, boundedness and the semigroup law give

$$\mathcal{E}_s(O) = \mathcal{E}_s\Big(\lim_{t\to\infty} \mathcal{E}_t(O)\Big) = \lim_{t\to\infty} \mathcal{E}_{s+t}(O) = O.$$

The finite-window quantity d_τ is used only when an honestly approximate notion is intended. Exact fixation remains an idealization of approximate einselection [9, 10].

Let

$$V_{\text{fg}}(\mathfrak{F}) = \operatorname{span}\{f_1, \ldots, f_{k^*}\}$$

be an orthonormal foreground basis selected by the T-DOME II rate-distortion optimum. Guided by the pointer basis selected in the broken phase [5], choose self-adjoint ambient observables

$$O_1, \ldots, O_{k^*} \in \mathcal{A}_U^{\text{sa}}$$

and define

$$\iota_{\mathfrak{F}}(f_i) := O_i, \qquad \iota_{\mathfrak{F}}\Big(\sum_{i=1}^{k^*} a_i f_i\Big) := \sum_{i=1}^{k^*} a_i O_i.$$

Write

$$\mathfrak{S}_{\mathfrak{F}} := \mathrm{span}\{1, O_1, \ldots, O_{k^*}\} \subset \mathcal{A}_U$$

for the resulting operator system and

$$\mathcal{A}_{\mathfrak{F}} := W^*(\mathfrak{S}_{\mathfrak{F}})$$

for its von Neumann envelope.

Proposition 5.3 (Pointer realization of the foreground sector)**.** *Let $\mathfrak{F}$ be a T-DOME foreground frame with*

$$V_{\mathrm{fg}}(\mathfrak{F}) = \mathrm{span}\{f_1, \ldots, f_{k^*}\}.$$

Assume there exists a family of self-adjoint ambient observables

$$O_1, \ldots, O_{k^*} \in \mathcal{A}_U^{\mathrm{sa}}$$

such that:

1. *$O_1, \ldots, O_{k^*}$ are linearly independent;*
2. *each O_i is pointer-aligned under $\{\mathcal{E}_t\}$;*
3. *the ordering of their survival relevance matches the ordering selected by the T-DOME foreground optimization.*

Then the linear extension $f_i \mapsto O_i$ defines an injective map

$$\iota_{\mathfrak{F}} : V_{\mathrm{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\mathrm{sa}}.$$

Moreover,

$$\mathfrak{S}_{\mathfrak{F}} = \mathrm{span}\{1, O_1, \ldots, O_{k^*}\}$$

is a finite-dimensional operator system, and

$$\mathcal{A}_{\mathfrak{F}} = W^*(\mathfrak{S}_{\mathfrak{F}})$$

is a von Neumann subalgebra of $\mathcal{A}_U$.

Proof. Linear independence of $O_1, \ldots, O_{k^*}$ makes the linear extension of $f_i \mapsto O_i$ injective. Since the O_i are self-adjoint, $\mathfrak{S}_{\mathfrak{F}}$ is self-adjoint and contains the identity, hence is an operator system. By definition, $W^*(\mathfrak{S}_{\mathfrak{F}})$ is the von Neumann algebra generated by that operator system inside $\mathcal{A}_U$. Therefore $\mathcal{A}_{\mathfrak{F}}$ is a von Neumann subalgebra of $\mathcal{A}_U$. □

The next refinement is to replace the bare existence assumption on the ambient observables O_i by a more structured compatibility package that points directly back to T-DOME I and II.

The PS1–PS3 package. The structured foreground-realization input naturally separates into three pieces:

(PS1) in the exact dephasing-dominated regime, the decoherence semigroup admits a commuting exact fixed-point pointer sector inside $\mathcal{A}_U$;

(PS2) the T-DOME sufficient-statistic coordinates agree with the coordinates induced by that pointer sector;

(PS3) the T-DOME rate-distortion optimum selects an ordered subset of pointer channels by survival relevance.

In this decomposition, PS1 is an exact-regime idealization of the decoherence-side input (not a consequence of generic decoherence or quantum Darwinism alone, but of the noiseless-sector limit in which pointer projections are exactly fixed rather than approximately einselected), whereas PS2 is the genuine model-independent bridge bottleneck and PS3 is downstream from it.

Proposition 5.4 (Conditional pointer-sector realization)**.** *Assume:*

(PS1) in the exact dephasing-dominated regime, the decoherence semigroup admits a commuting family of nonzero exact fixed-point projections $\{P_1, \dots, P_D\} \subset \mathrm{Fix}(\mathcal{E})$ *generating a classical pointer sector*

$$\mathcal{A}_{\mathrm{ptr}} := W^*(P_1, \dots, P_D) \subseteq \mathcal{A}_U;$$

(PS2) the sufficient-statistic record coordinates used in T-DOME II are the coordinates induced by this pointer sector through the T-DOME I coarse-graining map to the classical sector;

(PS3) the T-DOME II rate-distortion optimum selects an ordered subset $I^* = \{i_1, \dots, i_{k^*}\} \subseteq \{1, \dots, D\}$ *by survival relevance.*

Let

$$V_{\mathrm{fg}}(\mathfrak{F}) = \mathrm{span}\{f_1, \dots, f_{k^*}\}$$

be ordered compatibly with I^**, and define*

$$\iota_{\mathfrak{F}}(f_j) := P_{i_j}.$$

Then:

1. $\iota_{\mathfrak{F}} : V_{\mathrm{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\mathrm{sa}}$ *is injective;*
2. *the image lies in the exact decoherence fixed-point algebra;*
3. *the induced pointer-generated algebra satisfies*

$$\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\mathrm{ptr}}.$$

Proof. By assumption (1), each P_i is self-adjoint and satisfies $\mathcal{E}_t(P_i) = P_i$ for all $t \geq 0$. Distinct nonzero orthogonal projections are linearly independent, so the assignment $f_j \mapsto P_{i_j}$ extends to an injective linear map $\iota_{\mathfrak{F}}$. Exact fixedness gives the second claim. Since the P_{i_j} lie in the commuting pointer sector $\mathcal{A}_{\mathrm{ptr}}$, the von Neumann algebra generated by $1, P_{i_1}, \dots, P_{i_{k^*}}$ is contained in $\mathcal{A}_{\mathrm{ptr}}$. □

Remark 5.5 (Status of PS1). PS1 is the least problematic part of the package. In decoherence theory and quantum Darwinism, robust classical records are carried by a commuting pointer sector selected by the environment [9, 10]. In the exact fixed-point regime used here, this is precisely the statement that the decoherence semigroup admits a commuting family of exact fixed-point projections generating a classical pointer algebra. HAFF Paper D realizes this exactly in the dephasing-dominated finite-dimensional model, where the diagonal fixed-point algebra is the pointer basis algebra [3]. The real model-independent unresolved content is therefore PS2, with PS3 following once PS2 is established.

The PS2 alignment problem. Proposition 5.4 sharpens the existence assumption on $\iota_{\mathfrak{F}}$, but it does not yet derive the compatibility package from the broken-phase output itself in full generality. Section 6 closes a first restricted version in a product-classical pointer regime. The real unresolved content is therefore the model-independent form of the middle condition: the claim that the T-DOME sufficient-statistic coordinates and the pointer-sector coordinates are the same optimization problem written in two languages. That is the full PS2 problem.

Target lemma (PS2 alignment; open). Let

$$\mathcal{A}_{\text{ptr}} = W^*(P_1, \ldots, P_D)$$

be the pointer sector from Proposition 5.4, and define the pointer-coordinate process

$$p_i(t) := \omega_t(P_i), \qquad i = 1, \ldots, D.$$

For an ordered subset

$$I = \{i_1, \ldots, i_k\} \subseteq \{1, \ldots, D\},$$

write $\mathcal{F}_I$ for the pointer-adapted frame retaining those channels. A natural target theorem is the following.

Assume:

1. the T-DOME sufficient statistics $\{c_i(t)\}_{i=1}^D$ agree, after reindexing, with the pointer-coordinate process $\{p_i(t)\}_{i=1}^D$;
2. there exist nonnegative weights $w_1, \ldots, w_D$ and a constant D_0 such that for every pointer-adapted frame $\mathcal{F}_I$,
$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i;$$
3. admissible frames satisfy $|I| = k^* = \lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$.

Then every survival-distortion minimizer retains exactly the k^* channels of largest weight, up to ties. In particular, the T-DOME foreground directions coincide with the top-ranked pointer directions, and the ordered subset I^* of Proposition 5.4 is selected canonically.

Proof strategy. This target should be attacked by an exchange argument. If an admissible subset I omits some channel a and retains some channel b with $w_a > w_b$, then replacing b by a changes the distortion by

$$D(\mathcal{F}_{I\setminus\{b\}\cup\{a\}}) - D(\mathcal{F}_I) = -w_a + w_b < 0,$$

contradicting optimality. Once the distortion has been reduced to such a channelwise omission law, alignment is immediate. The real analytic burden is therefore to derive the weights w_i from the T-DOME survival functional in a way compatible with the pointer-sector coarse-graining.

First realistic route: approximate PS2b in the decoherence-dominated regime. The exact separability law above is likely too strong as a first target in the fully correlated setting. A more plausible theorem program is to prove approximate separability once decoherence has suppressed the cross-channel terms strongly enough that the pointer sector is approximately classical.

Introduce a decoherence-depth parameter τ_{dec}, and let $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \to 0$ measure the size of the residual cross-channel distortion after coarse-graining beyond that scale. The natural asymptotic target is:

$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i + R_I, \qquad |R_I| \leq \varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Under this bound, the same exchange argument gives

$$D(\mathcal{F}_{I\setminus\{b\}\cup\{a\}}) - D(\mathcal{F}_I) \leq -(w_a - w_b) + 2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Therefore any exchange across a weight gap larger than $2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$ strictly improves the objective. In particular, if the threshold gap

$$\Delta_{k^*} := w_{(k^*)} - w_{(k^*+1)}$$

exceeds $2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$, then the top-$k^*$ pointer subset is forced, up to ties above the gap.

This is likely the right first Bridge 1 theorem target. It matches the physics of decoherence, which suppresses off-diagonal couplings rather than eliminating them exactly in full generality. Section 6 closes this target in a first restricted product-classical regime. The real analytic task is thus to prove that the T-DOME survival distortion admits such a uniform remainder estimate in the decoherence-dominated regime.

Analytic source of $\varepsilon_{\mathrm{dec}}$. The remainder estimate cannot come from abstract operator algebra alone. It has to be extracted from the concrete T-DOME I survival identity

$$\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}} - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle.$$

In the pointer basis, the natural next step is to decompose the record process and the induced state into diagonal and off-diagonal parts,

$$X(t) = X^{\mathrm{diag}}(t) + X^{\mathrm{off}}(t), \qquad \rho(t) = \rho_{\mathrm{diag}}(t) + \rho_{\mathrm{off}}(t),$$

where the diagonal part is generated by the commuting pointer projections and the off-diagonal part carries the residual coherences.

The target analytic lemma is then the following: in the decoherence-dominated regime, the off-diagonal contribution to the survival identity is uniformly bounded by a function $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \to 0$, so that for every pointer-adapted frame $\mathcal{F}_I$,

$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i + R_I, \qquad |R_I| \leq \varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

The likely proof route is:

1. use the decoherence semigroup to obtain exponential or semigroup-gap suppression of the off-pointer sector;
2. propagate that suppression through the three terms in the survival identity;
3. identify the diagonal leading term as an additive pointer-channel omission loss.

This identifies the analytic content that Section 6 closes in a first restricted regime.

6 A Restricted PS2 Result in the Decoherence-Dominated Pointer Regime

The full PS2 alignment problem remains open. Still, a genuine theorem can already be proved in a restricted regime. The point of the present section is not to claim a model-independent PS2 theorem, but to show that in a product-classical, decoherence-dominated pointer regime, the structural reduction to fragment-local omission losses and the estimate reduction to a uniformly small coherence remainder can both be closed explicitly, after which approximate separability and top-k^* selection follow by a completely explicit argument.

Proposition 6.1 (Restricted approximate PS2b in the product-classical pointer regime)**.** *Fix $k \in \{1, \dots, D-1\}$, and let $\mathcal{F}_I$ range over all pointer-adapted frames with $|I| = k$. Assume the T-DOME survival distortion uses the squared loss*

$$\ell(x) = x^2.$$

Suppose that for each admissible I there exist random variables

$$L_i = L_i(\xi), \qquad i = 1, \dots, D,$$

and a coherence remainder $C_I = C_I(\xi)$ such that

$$\mathcal{S}_{\text{full}}(\xi) - \mathcal{S}_{\mathcal{F}_I}(\xi) = \sum_{i \notin I} L_i(\xi) + C_I(\xi)$$

for almost every environmental realization ξ. Assume further:

1. ***Pairwise diagonal orthogonality:***

$$\mathbb{E}_\xi[L_i L_j] = 0 \qquad \textit{for } i \neq j;$$

2. ***Uniform coherence bound:***

$$\|C_I\|_{L^\infty(\xi)} \leq \eta_{\text{dec}}(\tau_{\text{dec}}) \qquad \textit{for every admissible } I;$$

3. ***Finite diagonal size:***

$$\Lambda_k := \sup_{|I|=k} \sum_{i \notin I} \|L_i\|_{L^2(\xi)} < \infty.$$

Define

$$w_i := \|L_i\|^2_{L^2(\xi)}$$

and

$$\varepsilon_{\text{dec}}(\tau_{\text{dec}}) := 2\Lambda_k\, \eta_{\text{dec}}(\tau_{\text{dec}}) + \eta_{\text{dec}}(\tau_{\text{dec}})^2.$$

Then for every admissible I,

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I, \qquad |R_I| \leq \varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

Consequently, if

$$\Delta_k := w_{(k)} - w_{(k+1)} > 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}),$$

then every minimizer of $D(\mathcal{F}_I)$ retains exactly the top-k pointer channels, up to ties above the gap.

Proof. By definition of the distortion and the squared loss,

$$D(\mathcal{F}_I) = \mathbb{E}_\xi \left[\left(\sum_{i \notin I} L_i + C_I \right)^2 \right].$$

Expanding the square gives

$$D(\mathcal{F}_I) = \sum_{i \notin I} \mathbb{E}_\xi[L_i^2] + 2 \sum_{\substack{i,j \notin I \\ i<j}} \mathbb{E}_\xi[L_i L_j] + 2 \sum_{i \notin I} \mathbb{E}_\xi[L_i C_I] + \mathbb{E}_\xi[C_I^2].$$

The pairwise orthogonality assumption cancels the middle sum, so

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I,$$

where

$$R_I := 2 \sum_{i \notin I} \mathbb{E}_\xi[L_i C_I] + \mathbb{E}_\xi[C_I^2].$$

Because the environmental expectation is taken against a probability measure, $\|C_I\|_{L^2(\xi)} \leq \|C_I\|_{L^\infty(\xi)}$. Hence

$$|\mathbb{E}_\xi[L_i C_I]| \leq \|L_i\|_{L^2(\xi)} \|C_I\|_{L^2(\xi)} \leq \|L_i\|_{L^2(\xi)} \eta_{\text{dec}}(\tau_{\text{dec}}),$$

and therefore

$$|R_I| \leq 2\eta_{\text{dec}}(\tau_{\text{dec}}) \sum_{i \notin I} \|L_i\|_{L^2(\xi)} + \eta_{\text{dec}}(\tau_{\text{dec}})^2 \leq \varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

This proves the approximate separability law.

Now let I be admissible, and suppose $a \notin I$, $b \in I$ with $w_a > w_b$. Writing

$$I' := I \setminus \{b\} \cup \{a\},$$

the additive part changes by

$$\sum_{i \notin I'} w_i - \sum_{i \notin I} w_i = -w_a + w_b.$$

The remainder terms satisfy

$$R_{I'} - R_I \leq |R_{I'}| + |R_I| \leq 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}),$$

so

$$D(\mathcal{F}_{I'}) - D(\mathcal{F}_I) \leq -(w_a - w_b) + 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

Therefore any exchange across a weight gap strictly larger than $2\varepsilon_{\text{dec}}(\tau_{\text{dec}})$ lowers the distortion. If $\Delta_k > 2\varepsilon_{\text{dec}}(\tau_{\text{dec}})$, no minimizer can omit a channel above the threshold while retaining one below it. Hence every minimizer is exactly the top-k pointer subset, up to ties above the gap. □

Lemma 6.2 (Product-classical coarse-graining yields fragment-local omission losses)**.** *Assume that after applying the pointer coarse-graining map Φ, the diagonal classical sector is described by D local channel records*

$$\xi = (\xi_1, \ldots, \xi_D), \qquad \xi_i = (z_i, e_i),$$

with full diagonal law

$$P^{\text{full}}(\xi) = \prod_{i=1}^{D} P_i(z_i, e_i).$$

Let $\mathcal{F}_I$ be an admissible pointer-adapted frame. Assume its diagonal projected law is obtained by retaining the full local channel law on I and replacing each omitted channel by a fixed reference law:

$$P^I(\xi) = \prod_{i \in I} P_i(z_i, e_i) \prod_{i \notin I} Q_i(z_i, e_i).$$

Assume further:

1. *the diagonal survival identity is represented pointwise by*

$$\beta\, \mathcal{S}^{\text{diag}}(\xi) = -\imath(\xi) - d(\xi) - \beta\, \delta h(\xi),$$

   where $\imath$ is the pointwise mutual-information density, d is the pointwise reference-divergence density, and δh is the pointwise control contribution;

2. *the reference divergence is taken with respect to a product reference law*

$$\tau(e) = \prod_{i=1}^{D} \tau_i(e_i);$$

3. *the diagonal control contribution is additive:*

$$\delta h^{\text{full}}(\xi) = \sum_{i=1}^{D} \delta h_i^{\text{full}}(\xi_i),$$

$$\delta h^{I}(\xi) = \sum_{i \in I} \delta h_i^{\text{full}}(\xi_i) + \sum_{i \notin I} \delta h_i^{\text{ref}}(\xi_i).$$

For each channel i, define the local pointwise mutual-information density

$$\imath_i^{\text{full}}(\xi_i) := \log \frac{P_i(z_i, e_i)}{P_i^Z(z_i)\, P_i^E(e_i)}, \qquad \imath_i^{\text{ref}}(\xi_i) := \log \frac{Q_i(z_i, e_i)}{Q_i^Z(z_i)\, Q_i^E(e_i)},$$

and the local reference-divergence density

$$d_i^{\text{full}}(e_i) := \log \frac{P_i^E(e_i)}{\tau_i(e_i)}, \qquad d_i^{\text{ref}}(e_i) := \log \frac{Q_i^E(e_i)}{\tau_i(e_i)}.$$

Then the diagonal survival deficit is fragment-local:

$$\beta\big(\mathcal{S}_{\text{full}}^{\text{diag}}(\xi) - \mathcal{S}_{\mathcal{F}_I}^{\text{diag}}(\xi)\big) = \sum_{i \notin I} \lambda_i(\xi_i),$$

where

$$\lambda_i(\xi_i) := -\big(\imath_i^{\text{full}}(\xi_i) - \imath_i^{\text{ref}}(\xi_i)\big) - \big(d_i^{\text{full}}(e_i) - d_i^{\text{ref}}(e_i)\big) - \beta\big(\delta h_i^{\text{full}}(\xi_i) - \delta h_i^{\text{ref}}(\xi_i)\big).$$

Consequently, the centered variables

$$L_i(\xi_i) := \lambda_i(\xi_i) - \mathbb{E}_\xi[\lambda_i(\xi_i)]$$

are fragment-local omission losses in the sense needed by Lemma 6.4.

Proof. Because the full diagonal law factorizes,

$$P^{\text{full}}(\xi) = \prod_{i=1}^{D} P_i(z_i, e_i), \quad P_Z^{\text{full}}(z) = \prod_{i=1}^{D} P_i^Z(z_i), \quad P_E^{\text{full}}(e) = \prod_{i=1}^{D} P_i^E(e_i).$$

Therefore the full pointwise mutual-information density is additive:

$$\imath^{\text{full}}(\xi) := \log \frac{P^{\text{full}}(\xi)}{P_Z^{\text{full}}(z)\, P_E^{\text{full}}(e)} = \sum_{i=1}^{D} \imath_i^{\text{full}}(\xi_i).$$

The same argument for the projected law gives

$$\imath^I(\xi) = \sum_{i\in I} \imath_i^{\text{full}}(\xi_i) + \sum_{i\notin I} \imath_i^{\text{ref}}(\xi_i).$$

Since the reference law τ is also a product law,

$$d^{\text{full}}(e) := \log \frac{P_E^{\text{full}}(e)}{\tau(e)} = \sum_{i=1}^{D} d_i^{\text{full}}(e_i),$$

and similarly

$$d^I(e) = \sum_{i\in I} d_i^{\text{full}}(e_i) + \sum_{i\notin I} d_i^{\text{ref}}(e_i).$$

The control contribution is additive by assumption:

$$\delta h^{\text{full}}(\xi) = \sum_{i=1}^{D} \delta h_i^{\text{full}}(\xi_i),$$

$$\delta h^I(\xi) = \sum_{i\in I} \delta h_i^{\text{full}}(\xi_i) + \sum_{i\notin I} \delta h_i^{\text{ref}}(\xi_i).$$

Substituting these three decompositions into the pointwise diagonal survival identity gives

$$\begin{aligned}
&\beta\big(\mathcal{S}_{\text{full}}^{\text{diag}}(\xi) - \mathcal{S}_{\mathcal{F}_I}^{\text{diag}}(\xi)\big) \\
&= -\big(\imath^{\text{full}}(\xi) - \imath^I(\xi)\big) - \big(d^{\text{full}}(e) - d^I(e)\big) - \beta\big(\delta h^{\text{full}}(\xi) - \delta h^I(\xi)\big) \\
&= \sum_{i\notin I}\Big[-\big(\imath_i^{\text{full}}(\xi_i) - \imath_i^{\text{ref}}(\xi_i)\big) - \big(d_i^{\text{full}}(e_i) - d_i^{\text{ref}}(e_i)\big) - \beta\big(\delta h_i^{\text{full}}(\xi_i) - \delta h_i^{\text{ref}}(\xi_i)\big)\Big].
\end{aligned}$$

This is exactly the stated fragment-local decomposition. Centering each λ_i produces the centered fragment-local omission losses L_i. □

Remark 6.3 (What Lemma 6.2 closes). Lemma 6.2 is the first theorem-level closure of the structure line PS2a. It does not derive the product-classical pointer regime from the broken-phase output; it shows that once the pointer coarse-graining reaches that regime, fragment-local omission losses follow automatically by additivity of information densities, reference-divergence densities, and diagonal control terms.

Lemma 6.4 (Orthogonality from fragment-local centered channel losses)**.** *Assume the environmental realization admits a pointer-channel factorization*

$$\xi = (\xi_1, \dots, \xi_D)$$

with independent coordinates. Suppose the diagonal omission losses of Proposition 6.1 admit a centered fragment-local realization

$$L_i(\xi) = \lambda_i(\xi_i) - \mathbb{E}_\xi[\lambda_i(\xi_i)], \qquad i = 1, \dots, D,$$

for measurable functions $\lambda_i(\xi_i) \in L^2(\xi_i)$. Then

$$\mathbb{E}_\xi[L_i L_j] = 0 \qquad \textit{for } i \neq j.$$

In particular, the pairwise orthogonality hypothesis of Proposition 6.1 is automatic under independent pointer channels together with centered fragment-local omission losses.

Proof. Because L_i depends only on ξ_i and L_j depends only on ξ_j, the independence of ξ_i and ξ_j implies

$$\mathbb{E}_\xi[L_i L_j] = \mathbb{E}_\xi[L_i]\, \mathbb{E}_\xi[L_j].$$

The centered form of L_i and L_j gives

$$\mathbb{E}_\xi[L_i] = 0 = \mathbb{E}_\xi[L_j],$$

hence $\mathbb{E}_\xi[L_i L_j] = 0$. □

Remark 6.5 (Why the orthogonality hypothesis is natural in the pointer regime). Lemma 6.4 explains why the first hypothesis of Proposition 6.1 is not an arbitrary add-on. On the T-DOME side, the computational-ceiling analysis already treats the active channels as independent rate carriers. On the HAFF/Quantum Darwinism side, accessible classical records are carried by independent environmental fragments. Once the diagonal omission loss localizes to fragment-wise centered channel contributions after coarse-graining, pairwise orthogonality becomes a probability identity rather than an extra dynamical postulate.

What remains open is exactly the next step: deriving that fragment-local centered realization, together with the uniform coherence bound, from the full T-DOME survival identity in a model-independent way.

Lemma 6.6 (Semigroup-decay coherence estimate)**.** *Let*

$$\mathbb{E}_{\text{ptr}} : \mathcal{A}_U \to \mathcal{A}_{\text{ptr}}$$

be a normal conditional expectation onto the pointer algebra, and write

$$Q_{\text{off}} := \text{id} - \mathbb{E}_{\text{ptr}}.$$

For each admissible pointer-adapted frame $\mathcal{F}_I$ and environmental realization ξ, let X_ξ denote the relevant state or record datum entering the survival identity, and decompose the coherence remainder as

$$C_I(\xi) = C_I^{(I)}(\xi) + C_I^{(D)}(\xi) + C_I^{(H)}(\xi),$$

where the three terms correspond respectively to the off-diagonal contributions of

$$-\Delta I(S{:}E), \qquad -\Delta D_{\text{KL}}, \qquad -\beta\, \Delta\langle H_{\text{ctrl}}\rangle.$$

Assume:

1. ***Off-sector semigroup decay:*** *there exists a non-increasing function* $g : [0, \infty) \to [0, \infty)$ *with* $g(\tau) \to 0$ *such that*

$$\sup_{t \geq \tau} \|Q_{\text{off}} \mathcal{E}_t(X)\| \leq g(\tau) \, \|Q_{\text{off}} X\|$$

for every relevant datum X*;*

2. ***Uniform off-sector size:***

$$\sup_{\xi} \|Q_{\text{off}} X_\xi\| \leq B_{\text{off}} < \infty;$$

3. ***Termwise continuity bounds:*** *there exist constants* $L_I, L_D, L_H \geq 0$ *such that for every admissible* I *and every* ξ*,*

$$|C_I^{(I)}(\xi)| \leq L_I \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|,$$

$$|C_I^{(D)}(\xi)| \leq L_D \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|,$$

$$|C_I^{(H)}(\xi)| \leq L_H \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|.$$

Then for every admissible I*,*

$$\|C_I\|_{L^\infty(\xi)} \leq \eta_{\text{dec}}(\tau_{\text{dec}}) := (L_I + L_D + L_H) \, B_{\text{off}} \, g(\tau_{\text{dec}}).$$

In particular, the uniform coherence bound required in Proposition 6.1 holds with this choice of η_{dec}*.*

Proof. By the triangle inequality,

$$|C_I(\xi)| \leq |C_I^{(I)}(\xi)| + |C_I^{(D)}(\xi)| + |C_I^{(H)}(\xi)|.$$

Applying the termwise continuity bounds gives

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|.$$

The off-sector semigroup decay assumption then yields

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \, g(\tau_{\text{dec}}) \, \|Q_{\text{off}} X_\xi\|.$$

Using the uniform off-sector bound,

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \, B_{\text{off}} \, g(\tau_{\text{dec}}).$$

Taking the essential supremum over ξ gives the stated L^∞ bound. □

Remark 6.7 (What Lemma 6.6 closes). Lemma 6.6 is the abstract translation step from standard decoherence-semigroup suppression of the off-pointer sector to the exact η_{dec} object used in Proposition 6.1. It does not yet identify the best norm or derive the constants L_I, L_D, L_H from the full T-DOME survival identity; those remain the true analytic content.

In concrete pointer-diagonal control models, one often has $L_H = 0$, so only the mutual-information and relative-entropy terms contribute to the coherence remainder.

Corollary 6.8 (Hilbert–Schmidt continuity package on a uniformly faithful pointer support)**.** *Assume that, after pointer coarse-graining, the relevant joint system–environment states are supported on a finite-dimensional space*

$$\mathcal{H}_S \otimes \mathcal{H}_E$$

with

$$d_S := \dim \mathcal{H}_S, \qquad d_E := \dim \mathcal{H}_E.$$

Assume there exists $m_* > 0$ such that every relevant joint state ρ_{SE}, every relevant marginal state ρ_S, ρ_E, and the fixed reference state τ_E entering the T-DOME ΔD_{KL} term satisfy

$$\lambda_{\min}(\rho_{SE}), \lambda_{\min}(\rho_S), \lambda_{\min}(\rho_E), \lambda_{\min}(\tau_E) \geq m_*.$$

Write $\|\cdot\|_2$ for the Hilbert–Schmidt norm.
Then:

1. *for any two relevant joint states ρ_{SE}, σ_{SE},*

$$\big|I(\rho_{SE}) - I(\sigma_{SE})\big| \leq 3\sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\,\|\rho_{SE} - \sigma_{SE}\|_2;$$

2. *for any two relevant environment states ρ_E, σ_E,*

$$\big|D_{\mathrm{KL}}(\rho_E\|\tau_E) - D_{\mathrm{KL}}(\sigma_E\|\tau_E)\big| \leq \sqrt{d_E}\,\big(2|\log m_*| + 1\big)\,\|\rho_E - \sigma_E\|_2;$$

3. *for any self-adjoint control Hamiltonian $H_{\mathrm{ctrl}} \in \mathcal{B}(\mathcal{H}_S)$,*

$$\big|\mathrm{Tr}[H_{\mathrm{ctrl}}(\rho_S - \sigma_S)]\big| \leq \|H_{\mathrm{ctrl}}\|_2\,\|\rho_S - \sigma_S\|_2.$$

Consequently, in this restricted finite-dimensional faithful regime, one may take Hilbert–Schmidt continuity constants

$$L_I^{\mathrm{HS}} := 3\sqrt{d_S d_E}\,\big(|\log m_*| + 1\big),$$

$$L_D^{\mathrm{HS}} := \sqrt{d_E}\,\big(2|\log m_*| + 1\big),$$

$$L_H^{\mathrm{HS}} := \|H_{\mathrm{ctrl}}\|_2$$

in Lemma 6.6, after translating the relevant off-sector perturbations into the corresponding Hilbert–Schmidt norms.

Proof. Set

$$S(\rho) := -\,\mathrm{Tr}(\rho \log \rho).$$

On the convex set of density operators with spectrum contained in $[m_*, 1]$, the Fréchet derivative is

$$DS_\rho(H) = -\,\mathrm{Tr}\big(H(\log\rho + \mathbf{1})\big).$$

Hence

$$|DS_\rho(H)| \leq \|H\|_2\,\|\log\rho + \mathbf{1}\|_2 \leq \sqrt{d}\,\big(|\log m_*| + 1\big)\|H\|_2$$

for $d = \dim \mathrm{supp}\,\rho$. Applying the mean-value theorem along the line segment between two states gives

$$|S(\rho) - S(\sigma)| \leq \sqrt{d}\,\big(|\log m_*| + 1\big)\|\rho - \sigma\|_2.$$

For the mutual-information term, use

$$I(\rho_{SE}) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}).$$

The partial trace satisfies the Hilbert–Schmidt bound

$$\|\operatorname{Tr}_E A\|_2 \le \sqrt{d_E}\,\|A\|_2, \qquad \|\operatorname{Tr}_S A\|_2 \le \sqrt{d_S}\,\|A\|_2.$$

Therefore

$$|S(\rho_S) - S(\sigma_S)| \le \sqrt{d_S}\,\big(|\log m_*| + 1\big)\|\rho_S - \sigma_S\|_2 \le \sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\|\rho_{SE} - \sigma_{SE}\|_2,$$

and similarly for the environment marginal. For the joint entropy term,

$$|S(\rho_{SE}) - S(\sigma_{SE})| \le \sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\|\rho_{SE} - \sigma_{SE}\|_2.$$

Summing the three contributions proves part (1).

For the relative-entropy term with fixed faithful reference τ_E, write

$$D_{\mathrm{KL}}(\rho_E\|\tau_E) = \operatorname{Tr}(\rho_E \log \rho_E) - \operatorname{Tr}(\rho_E \log \tau_E).$$

Hence

$$\begin{aligned}&\big|D_{\mathrm{KL}}(\rho_E\|\tau_E) - D_{\mathrm{KL}}(\sigma_E\|\tau_E)\big| \\ &\le \big|S(\rho_E) - S(\sigma_E)\big| + \big|\operatorname{Tr}\big((\rho_E - \sigma_E)\log\tau_E\big)\big|.\end{aligned}$$

The entropy term is bounded by

$$\sqrt{d_E}\,\big(|\log m_*| + 1\big)\|\rho_E - \sigma_E\|_2,$$

while

$$\|\log \tau_E\|_2 \le \sqrt{d_E}\,|\log m_*|.$$

Cauchy–Schwarz therefore gives part (2).

Part (3) is immediate from the Hilbert–Schmidt Cauchy–Schwarz inequality. □

Remark 6.9 (Status of Corollary 6.8). Corollary 6.8 does not close the full estimate track in general Type III form. It closes the first finite-dimensional faithful pointer-support regime, where the Hilbert–Schmidt norm is natural and all three continuity constants become explicit. For the restricted Bridge 1 target developed in the present note, that is enough: the estimate line is closed at theorem level. The further question of whether an equally clean package survives beyond that regime, or must be reformulated in a predual norm, belongs to a later strengthening rather than to the current closure claim.

Corollary 6.10 (Restricted PS2 closure in the product-classical faithful pointer regime)**.** *Assume:*

1. *the hypotheses of Lemma 6.2;*
2. *the independent pointer-channel factorization hypothesis of Lemma 6.4;*
3. *the off-sector semigroup decay and uniform off-sector size hypotheses of Lemma 6.6;*
4. *the finite-dimensional uniformly faithful pointer-support hypotheses of Corollary 6.8;*

5. *the centered fragment-local omission losses from Lemma 6.2 satisfy*

$$\Lambda_k := \sup_{|I|=k} \sum_{i \notin I} \|L_i\|_{L^2(\xi)} < \infty.$$

Define

$$w_i := \|L_i\|^2_{L^2(\xi)},$$
$$\eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) := \left(L^{\mathrm{HS}}_I + L^{\mathrm{HS}}_D + L^{\mathrm{HS}}_H\right) B_{\mathrm{off}}\, g(\tau_{\mathrm{dec}}),$$

and

$$\varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) := 2\Lambda_k\, \eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) + \left(\eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}})\right)^2.$$

Then every admissible pointer-adapted frame $\mathcal{F}_I$ satisfies

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I, \qquad |R_I| \le \varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Consequently, if

$$\Delta_k := w_{(k)} - w_{(k+1)} > 2\varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}),$$

then every minimizer of $D(\mathcal{F}_I)$ retains exactly the top-k pointer channels, up to ties above the gap.

Proof. Lemma 6.2 produces centered fragment-local omission losses L_i. Lemma 6.4 upgrades the independent pointer-channel factorization to the pairwise orthogonality required in Proposition 6.1. By Corollary 6.8, the continuity constants appearing in Lemma 6.6 may be chosen as $L^{\mathrm{HS}}_I, L^{\mathrm{HS}}_D, L^{\mathrm{HS}}_H$, so that Lemma 6.6 yields the uniform coherence bound

$$\|C_I\|_{L^\infty(\xi)} \le \eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

The finiteness hypothesis gives the remaining size bound $\Lambda_k < \infty$. Proposition 6.1 therefore applies and gives exactly the stated approximate separability law and top-k selection criterion. □

Corollary 6.11 (Pure-dephasing envelope)**.** *Assume the hypotheses of Proposition 6.1, and suppose in addition that the coherence remainder is controlled by the omitted pointer-channel coherences in the form*

$$|C_I(\xi, t)| \le \sum_{i \notin I} a_i\, |p_i(t)| \qquad \textit{for all } t \ge \tau_{\mathrm{dec}},$$

where $a_i \ge 0$ are constants and each $p_i(t)$ is the exact Lorentz–Drude pure-dephasing decoherence function from T-DOME I [4, 7],

$$\gamma_i < 4\lambda_i, \qquad \Omega_i := \tfrac{1}{2}\sqrt{4\lambda_i\gamma_i - \gamma_i^2} \in \mathbb{R}_{>0},$$

and

$$p_i(t) = e^{-\gamma_i t/2}\left[\cos(\Omega_i t) + \frac{\gamma_i}{2\Omega_i}\sin(\Omega_i t)\right].$$

Define

$$R_i := \sqrt{1 + \left(\frac{\gamma_i}{2\Omega_i}\right)^2}, \qquad A_k := \max_{|I|=k} \sum_{i \notin I} a_i R_i.$$

Then one may take

$$\eta_{\mathrm{dec}}(\tau_{\mathrm{dec}}) = A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2}, \qquad \gamma_* := \min_{1\le i\le D} \gamma_i,$$

and hence

$$\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \le 2\Lambda_k A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2} + A_k^2\, e^{-\gamma_* \tau_{\mathrm{dec}}}.$$

In particular, the distortion remainder decays exponentially with the decoherence depth.

Proof. For each channel,

$$|p_i(t)| \le e^{-\gamma_i t/2} \sqrt{1 + \left(\frac{\gamma_i}{2\Omega_i}\right)^2} = R_i\, e^{-\gamma_i t/2} \le R_i\, e^{-\gamma_* t/2}$$

because $|a\cos x + b \sin x| \le \sqrt{a^2+b^2}$. Therefore

$$|C_I(\xi,t)| \le \sum_{i\notin I} a_i R_i\, e^{-\gamma_* t/2} \le A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2}$$

for all $t \ge \tau_{\mathrm{dec}}$. This identifies a valid choice of $\eta_{\mathrm{dec}}(\tau_{\mathrm{dec}})$. The stated bound on $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$ is then immediate from Proposition 6.1. □

A bookkeeping split for the model-independent PS2 programme. The restricted results above close the PS2 package only after explicit regime assumptions have been imposed. For the later model-independent programme, one further lemma is useful already now: it fixes the exact place where the off-sector estimates enter the survival deficit.

For $X \in \{\mathrm{full}, \mathcal{F}_I\}$, write the survival identity as

$$\beta\, \mathcal{S}_X = -\Delta I_X - \Delta D_X - \beta\, \Delta H_X,$$

where

$$\Delta D_X := \Delta D_{\mathrm{KL},X}, \qquad \Delta H_X := \Delta \langle H_{\mathrm{ctrl}} \rangle_X.$$

Assume each of the three terms admits a pointer-diagonal/off-sector decomposition,

$$\Delta I_X := \Delta I_X^{\mathrm{diag}} + \Delta I_X^{\mathrm{off}},$$

$$\Delta D_X := \Delta D_X^{\mathrm{diag}} + \Delta D_X^{\mathrm{off}},$$

$$\Delta H_X := \Delta H_X^{\mathrm{diag}} + \Delta H_X^{\mathrm{off}}.$$

Define the pointer-diagonal survival functional and off-sector remainder by

$$\beta\, \mathcal{S}_X^{\mathrm{diag}} := -\Delta I_X^{\mathrm{diag}} - \Delta D_X^{\mathrm{diag}} - \beta\, \Delta H_X^{\mathrm{diag}},$$

$$R_X^{\mathrm{off}} := -\Delta I_X^{\mathrm{off}} - \Delta D_X^{\mathrm{off}} - \beta\, \Delta H_X^{\mathrm{off}}.$$

Lemma 6.12 (Diagonal/off-sector split of the survival deficit)**.** *Let $\mathbb{E}_{\mathrm{ptr}} : \mathcal{A}_U \to \mathcal{A}_{\mathrm{ptr}}$ be the conditional expectation onto the pointer algebra selected by the broken phase, and assume the above termwise decompositions are computed with respect to $\mathbb{E}_{\mathrm{ptr}}$. Then for every admissible pointer-adapted frame $\mathcal{F}_I$,*

$$\beta\big(\mathcal{S}_{\mathrm{full}} - \mathcal{S}_{\mathcal{F}_I}\big) = \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}} - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}\big) + R_I^{\mathrm{off}},$$

where

$$R_I^{\mathrm{off}} := R_{\mathrm{full}}^{\mathrm{off}} - R_{\mathcal{F}_I}^{\mathrm{off}}.$$

Equivalently,

$$\begin{aligned} R_I^{\mathrm{off}} = &- \big(\Delta I_{\mathrm{full}}^{\mathrm{off}} - \Delta I_{\mathcal{F}_I}^{\mathrm{off}}\big) \\ &- \big(\Delta D_{\mathrm{full}}^{\mathrm{off}} - \Delta D_{\mathcal{F}_I}^{\mathrm{off}}\big) \\ &- \beta\big(\Delta H_{\mathrm{full}}^{\mathrm{off}} - \Delta H_{\mathcal{F}_I}^{\mathrm{off}}\big). \end{aligned}$$

Proof. Write the survival identity separately for the full process and the frame-adapted process $\mathcal{F}_I$, insert the termwise diagonal/off-sector decompositions into both identities, and subtract. The pointer-diagonal contributions group into

$$\beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}} - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}\big),$$

while the off-sector contributions group into

$$R_{\mathrm{full}}^{\mathrm{off}} - R_{\mathcal{F}_I}^{\mathrm{off}}.$$

This is exactly the stated identity. □

Remark 6.13 (What this does and does not close). Section 6 is a restricted theorem-level closure of PS2, not a general one. What is now closed is the implication

$$\begin{aligned} \text{uniformly small coherence remainder} &\Longrightarrow \text{approximate separability} \\ &\Longrightarrow \text{top-}k \text{ pointer selection.} \end{aligned}$$

Corollary 6.10 packages this as a single restricted selection theorem in the product-classical and finite-dimensional faithful pointer regime, while Corollary 6.11 gives a model-specific exponential envelope. Lemma 6.12 does not enlarge that closure claim; it only fixes the bookkeeping split showing that the already-controlled semigroup-decay terms belong to the off-sector remainder. What remains open is the model-independent derivation of the regime assumptions, and in particular control of the diagonal product defect, from the full T-DOME survival identity and the broken-phase output.

A conditional model-independent foreground-background package. The bookkeeping split also makes it possible to state an honest conditional version of the next model-independent target. For budget-feasible I with $|I| = k^*$, define the diagonal-tail survival deficit by

$$\Delta_I^{\mathrm{diag}}(\xi, t) := \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}(\xi, t)\big),$$

the corresponding tail distortion by

$$D_\tau^{\mathrm{diag}}(I) := \sup_{t \geq \tau} \mathbb{E}_{P_t^{\mathrm{diag}}}\Big[\big(\Delta_I^{\mathrm{diag}}(\xi, t)\big)^2\Big],$$

and for an exchange pair

$$\mathfrak{P} := I^* \times (I^*)^{\mathrm{c}}, \qquad N_{\mathrm{ex}} := |\mathfrak{P}| = k^*(D - k^*),$$

define

$$\begin{aligned} G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi, t) &:= \beta\big(\mathcal{S}_{\mathcal{F}_{I_{a\leftrightarrow b}}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathcal{F}_{I^*}}^{\mathrm{diag}}(\xi, t)\big), \\ \gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi, t) &:= \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathrm{ff}}^{\mathrm{diag}}(u, t) - \mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(v, t)\big), \\ \overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t) &:= \mathbb{E}_{P_t^{\mathrm{diag}}}\big[\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi, t)\big]. \end{aligned}$$

Proposition 6.14 (Conditional diagonal foreground-background coupling suppression)**.** *Let I^* be the broken-phase foreground set. Assume:*

1. ***Approximate tail optimality.*** *There exists $r_{\mathrm{opt}}(\tau) \to 0$ such that*

$$D_\tau^{\mathrm{diag}}(I^*) \le D_\tau^{\mathrm{diag}}(I) + r_{\mathrm{opt}}(\tau) \qquad \textit{for all } |I| = k^*.$$

2. ***Exchange cover and tail persistence.*** *There exist $c_{\mathrm{cov}} > 0$, $M_\gamma < \infty$, $e_{\mathrm{cov}}(\tau) \to 0$, and $r_{\mathrm{pers}}(\tau) \to 0$ such that:*

$$G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t)\,\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t) \ge 0 \qquad \textit{for all } (a,b) \in \mathfrak{P},\ t \ge \tau,$$

$$\frac{1}{N_{\mathrm{ex}}} \sum_{(a,b)\in\mathfrak{P}} \left|G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t)\right| \ge c_{\mathrm{cov}} \left|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t)\right| - e_{\mathrm{cov}}(\tau) \quad \textit{for a.e. } \xi,\ t \ge \tau,$$

$$\sup_{t\ge\tau} \mathbb{E}\big[|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\cdot,t)|\big] \le M_\gamma,$$

and for

$$\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) := \mathbb{E}\big[G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\,\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\cdot,t)\big]$$

one has

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{s,t\ge\tau} \left|\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) - \Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(s)\right| \le r_{\mathrm{pers}}(\tau).$$

3. ***Tail-passive background survival.*** *There exist $M_{\mathrm{ex}}(\tau) < \infty$ and $\eta_{\mathrm{bb}}(\tau) \to 0$ such that*

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{t\ge\tau} \left\|G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\right\|_{L^1} \le M_{\mathrm{ex}}(\tau),$$

$$\sup_{t\ge\tau} \left\|\mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(\cdot,t)\right\|_{L^\infty} \le \eta_{\mathrm{bb}}(\tau).$$

4. ***Exchange efficiency gap.*** *There exist $M_G(\tau) < \infty$, $m_\gamma(\tau) > 0$, and a constant $\delta_Q > 0$ such that*

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{t\ge\tau} \left\|G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\right\|_{L^\infty} \le M_G(\tau),$$

$$\left|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t)\right| \ge m_\gamma(\tau) \quad \textit{whenever } G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t) \ne 0,$$

and the efficiency ratio

$$\kappa_Q(\tau) := \frac{M_G(\tau)}{m_\gamma(\tau)}$$

satisfies

$$\kappa_Q(\tau) \le 2 - \delta_Q \qquad \textit{for all sufficiently large } \tau.$$

Then there exists a function $\varepsilon_\Gamma(\tau) \to 0$ such that

$$\sup_{t\ge\tau} \left|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)\right| \le \varepsilon_\Gamma(\tau)$$

for all sufficiently large τ. One may take

$$\varepsilon_\Gamma(\tau) := \left[\frac{N_{\mathrm{ex}}}{\delta_Q c_{\mathrm{cov}}} \left(r_{\mathrm{opt}}(\tau) + \frac{2e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma + 2r_{\mathrm{pers}}(\tau) + 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau)\right)\right]^{1/2}.$$

Proof. The exact exchange expansion gives, for every exchange pair,

$$D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*) \le Q_{a\leftrightarrow b}^\tau + 2B_{a\leftrightarrow b}^\tau - 2A_{a\leftrightarrow b}^\tau,$$

where

$$Q_{a\leftrightarrow b}^\tau := \sup_{t\ge\tau} \mathbb{E}\Big[\big(G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\big)^2\Big],$$

$$B_{a\leftrightarrow b}^\tau := \sup_{t\ge\tau} \Big|\beta\,\mathbb{E}\Big[G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\,\mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(\cdot,t)\Big]\Big|,$$

$$A_{a\leftrightarrow b}^\tau := \inf_{t\ge\tau} \Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t).$$

If

$$\sup_{t\ge\tau} \big|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)\big| > \varepsilon,$$

then the exchange-cover and tail-persistence assumptions imply the existence of $(a,b) \in \mathfrak{P}$ with

$$A_{a\leftrightarrow b}^\tau \ge \frac{c_{\mathrm{cov}}}{N_{\mathrm{ex}}}\,\varepsilon^2 - \frac{e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma - r_{\mathrm{pers}}(\tau).$$

The tail-passive background assumption gives

$$2B_{a\leftrightarrow b}^\tau \le 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau).$$

The exchange-efficiency gap implies

$$Q_{a\leftrightarrow b}^\tau \le (2-\delta_Q)\sup_{t\ge\tau}\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) \le (2-\delta_Q)\big(A_{a\leftrightarrow b}^\tau + r_{\mathrm{pers}}(\tau)\big).$$

Substituting these estimates into the exchange bound gives

$$D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*) \le -\frac{\delta_Q c_{\mathrm{cov}}}{N_{\mathrm{ex}}}\,\varepsilon^2 + \frac{2e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma + 2r_{\mathrm{pers}}(\tau) + 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau).$$

Approximate tail optimality of I^* yields

$$-r_{\mathrm{opt}}(\tau) \le D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*).$$

Combining the two inequalities gives the stated bound on ε, hence on $\sup_{t\ge\tau}|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)|$. □

Remark 6.15 (Compatibility and physical nonvacuity of Proposition 6.14). An engineered finite classical Markov witness constructed in the audit layer realizes the four tail packages simultaneously. It establishes their abstract mathematical compatibility, so the antecedent is not formally contradictory. It is not a natural source-datum model and is not used as a physical example here. No known physical source/control model currently derives all four packages jointly, and no known example also realizes the Type III modular shell interface needed downstream. Thus Proposition 6.14 remains a conditional result; its natural physical nonvacuity is open.

Corollary 6.16 (Conditional foreground-background mutual-information decay)**.** *Assume the hypotheses of Proposition 6.14, and in addition suppose that the pointwise diagonal foreground-background defect admits the block-local decomposition*

$$\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t) = -\imath_{\mathrm{fb}}(\xi,t) + e_{\mathrm{fb}}^D(\xi,t) + e_{\mathrm{fb}}^H(\xi,t),$$

where

$$\imath_{\mathrm{fb}}(\xi,t) := \log \frac{P_t^{\mathrm{diag}}(u,v)}{P_t^{\mathrm{diag}}(u)P_t^{\mathrm{diag}}(v)}$$

and the averaged locality defects satisfy

$$\sup_{t\geq\tau}\big|\overline{E}^{D}_{\mathrm{fb}}(t)\big| \leq \varepsilon_D(\tau), \qquad \sup_{t\geq\tau}\big|\overline{E}^{H}_{\mathrm{fb}}(t)\big| \leq \varepsilon_H(\tau),$$

with $\varepsilon_D(\tau), \varepsilon_H(\tau) \to 0$. *Then*

$$\sup_{t\geq\tau} I_t(u:v) \leq \varepsilon_\Gamma(\tau) + \varepsilon_D(\tau) + \varepsilon_H(\tau),$$

hence $I_t(u:v) \to 0$ *uniformly on the tail.*

Proof. Averaging the assumed pointwise decomposition gives

$$\overline{\Gamma}^{\mathrm{diag}}_{\mathrm{fb}}(t) = -I_t(u:v) + \overline{E}^{D}_{\mathrm{fb}}(t) + \overline{E}^{H}_{\mathrm{fb}}(t).$$

Taking absolute values and applying Proposition 6.14 gives the stated estimate. □

Corollary 6.17 (Conditional full diagonal product-defect closure)**.** *Fix orderings*

$$B_{\mathrm{fg}} = (i_1,\dots,i_{k^*}), \qquad B_{\mathrm{bg}} = (j_1,\dots,j_{D-k^*})$$

of the foreground and background indices. For an ordered block $B = (\ell_1,\dots,\ell_r)$*, write*

$$X_{B,\leq m} := (\xi_{\ell_1},\dots,\xi_{\ell_m}), \qquad I_{B,m}(t) := I_t\big(X_{B,\leq m} : \xi_{\ell_{m+1}}\big).$$

Assume the hypotheses of Corollary 6.16, and in addition suppose that for each ordered block $B \in \{B_{\mathrm{fg}}, B_{\mathrm{bg}}\}$ *and each* $m = 1,\dots,|B|-1$ *there is a pointwise diagonal prefix defect*

$$\gamma^{\mathrm{diag}}_{B,m}(x_{B,\leq m+1},t) = -\imath_{B,m}(x_{B,\leq m+1},t) + e^{D}_{B,m}(x_{B,\leq m+1},t) + e^{H}_{B,m}(x_{B,\leq m+1},t),$$

where

$$\imath_{B,m}(x_{B,\leq m+1},t) := \log \frac{P^{\leq m+1}_{B,t}(x_{B,\leq m+1})}{P^{\leq m}_{B,t}(x_{B,\leq m})\,P_{\ell_{m+1},t}(x_{\ell_{m+1}})},$$

and the averaged defects satisfy

$$\sup_{t\geq\tau}\big|\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)\big| \leq \varepsilon^{\Gamma}_{B,m}(\tau),$$

$$\sup_{t\geq\tau}\big|\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]\big| \leq \varepsilon^{D}_{B,m}(\tau), \qquad \sup_{t\geq\tau}\big|\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big]\big| \leq \varepsilon^{H}_{B,m}(\tau),$$

with all these functions tending to zero as $\tau\to\infty$. *Define*

$$\varepsilon^{\mathrm{seq}}_{\mathrm{fg}}(\tau) := \sum_{m=1}^{k^*-1}\Big(\varepsilon^{\Gamma}_{\mathrm{fg},m}(\tau) + \varepsilon^{D}_{\mathrm{fg},m}(\tau) + \varepsilon^{H}_{\mathrm{fg},m}(\tau)\Big),$$

$$\varepsilon^{\mathrm{seq}}_{\mathrm{bg}}(\tau) := \sum_{m=1}^{D-k^*-1}\Big(\varepsilon^{\Gamma}_{\mathrm{bg},m}(\tau) + \varepsilon^{D}_{\mathrm{bg},m}(\tau) + \varepsilon^{H}_{\mathrm{bg},m}(\tau)\Big),$$

with the convention that an empty sum is zero. Then

$$\sup_{t\geq\tau} D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}\middle\|\prod_{i=1}^{D}P_{i,t}\right) \leq \varepsilon_\Gamma(\tau) + \varepsilon_D(\tau) + \varepsilon_H(\tau) + \varepsilon^{\mathrm{seq}}_{\mathrm{fg}}(\tau) + \varepsilon^{\mathrm{seq}}_{\mathrm{bg}}(\tau).$$

Proof. The exact block decomposition gives

$$D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}\middle\|\prod_{i=1}^{D}P_{i,t}\right)=I_t(u:v)+D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}(u)\middle\|\prod_{i\in I^*}P_{i,t}\right)+D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}(v)\middle\|\prod_{j\notin I^*}P_{j,t}\right).$$

For any ordered block $B=(\ell_1,\dots,\ell_r)$, the KL chain rule gives

$$D_{\mathrm{KL}}\left(P_{B,t}\middle\|\prod_{q=1}^{r}P_{\ell_q,t}\right)=\sum_{m=1}^{r-1}I_{B,m}(t).$$

Averaging the assumed prefix decomposition yields

$$\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)=-I_{B,m}(t)+\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]+\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big],$$

hence

$$I_{B,m}(t)\le\left|\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)\right|+\left|\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]\right|+\left|\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big]\right|.$$

Summing over m gives the stated bound for each block. Combining those two blockwise bounds with Corollary 6.16 proves the claim. □

7 Stable-Hull Transfer and Comparison with the Exact Fixed Sector

Let $\omega(\cdot)=\langle\Omega,\cdot\,\Omega\rangle$ on $\mathcal{A}_U$, and let $\sigma_s^{\omega,\mathrm{tot}}$ denote the ambient modular flow. Define

$$\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}:=\left\{\begin{array}{l}\mathcal{M}\subseteq\mathcal{A}_U:\mathcal{M}\text{ is a von Neumann subalgebra,}\\ \qquad\mathcal{A}_{\mathfrak{F}}\subseteq\mathcal{M},\\ \qquad\sigma_s^{\omega,\mathrm{tot}}(\mathcal{M})=\mathcal{M}\quad\forall s\in\mathbb{R},\\ \qquad\mathcal{E}_t(\mathcal{M})\subseteq\mathcal{M}\quad\forall t\ge0\end{array}\right\}.$$

This class is nonempty because $\mathcal{A}_U\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$.

Three distinct structural inputs. For a general normal UCP semigroup, the linear fixed set

$$\mathrm{Fix}(\mathcal{E}):=\{A\in\mathcal{A}_U:\mathcal{E}_t(A)=A\ \forall t\ge0\}$$

is an operator system but need not be a von Neumann algebra under the ambient product. Whenever the comparison results below use

$$\mathcal{A}_{\mathrm{fix}}:=\mathrm{Fix}(\mathcal{E})$$

as an algebra, they assume separately:

(FSA) *fixed-set algebra closure:* $\mathcal{A}_{\mathrm{fix}}$ is a von Neumann subalgebra of $\mathcal{A}_U$;

(MOD) *Type III modular invariance:* $\sigma_s^{\omega,\mathrm{tot}}(\mathcal{A}_{\mathrm{fix}})=\mathcal{A}_{\mathrm{fix}}$ for every $s\in\mathbb{R}$;

(EXP) *expectedness when required:* there is a faithful normal ω-preserving conditional expectation $E_{\mathrm{fix}}:\mathcal{A}_U\to\mathcal{A}_{\mathrm{fix}}$.

FSA is a multiplicative-closure condition and is not supplied by UCP fixedness alone. MOD is a separate dynamical condition. In the present faithful standard-form setting, MOD permits the Takesaki expectation, but the mere existence of a faithful invariant state does not by itself establish MOD or EXP [12, 13]. The minimal stable-hull intersection below does not require EXP; an ω-preserving projection onto the exact fixed sector does.

Definition 7.1 (Stable-hull transfer map)**.** The *stable-hull transfer* of the foreground frame $\mathfrak{F}$ is

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) := \bigcap_{\mathcal{M}\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}} \mathcal{M}.$$

Proposition 7.2 (Preliminary inclusion in the exact fixed sector)**.** *Assume FSA. Thus $\mathcal{A}_{\mathrm{fix}} = \mathrm{Fix}(\mathcal{E})$ is a von Neumann subalgebra, without identifying it with probe accessibility or with an environmental record algebra. Let $\mathfrak{F}$ be a T-DOME foreground frame, and let $O_1, \dots, O_{k^*}$ be exact pointer-aligned observables realizing $\iota_{\mathfrak{F}}$ as in Proposition 5.3. Then*

$$\mathcal{A}_{\mathfrak{F}} = W^*(1, O_1, \dots, O_{k^*}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. Exact pointer alignment gives

$$O_i \in \mathrm{Fix}(\mathcal{E}) = \mathcal{A}_{\mathrm{fix}} \qquad \text{for each } i.$$

Since $\mathcal{A}_{\mathrm{fix}}$ is a von Neumann algebra by FSA, it contains the von Neumann algebra generated by $1, O_1, \dots, O_{k^*}$. Hence

$$W^*(1, O_1, \dots, O_{k^*}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

□

Proposition 7.3 (Minimal stable-hull transfer)**.** *Assume the hypotheses of Proposition 5.3. Then*

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) = \bigcap_{\mathcal{M}\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}} \mathcal{M}$$

is a well-defined von Neumann subalgebra of $\mathcal{A}_U$ satisfying:

1. $\mathcal{A}_{\mathfrak{F}} \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$*;*
2. $$\sigma^{\omega,\mathrm{tot}}_s\big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})\big) = \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \qquad \forall s \in \mathbb{R};$$
3. $$\mathcal{E}_t\big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})\big) \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \qquad \forall t \geq 0;$$
4. *it is minimal with these properties: if $\mathcal{N} \subseteq \mathcal{A}_U$ is any von Neumann subalgebra containing $\mathcal{A}_{\mathfrak{F}}$ and stable under both ambient dynamics, then*
$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{N}.$$

Proof. The class $\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$ is nonempty because $\mathcal{A}_U$ belongs to it. An arbitrary intersection of von Neumann subalgebras is again a von Neumann subalgebra, so the intersection is well-defined and is itself a von Neumann subalgebra of $\mathcal{A}_U$. Since every element of $\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$ contains $\mathcal{A}_{\mathfrak{F}}$, so does the intersection.

If $A \in \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$, then $A \in \mathcal{M}$ for every $\mathcal{M} \in \mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$. By definition of the class, $\sigma^{\omega,\mathrm{tot}}_s(A) \in \mathcal{M}$ for every $s \in \mathbb{R}$, and $\mathcal{E}_t(A) \in \mathcal{M}$ for every $t \geq 0$. Therefore both $\sigma^{\omega,\mathrm{tot}}_s(A)$ and $\mathcal{E}_t(A)$ belong to the intersection. This proves $\sigma^{\omega,\mathrm{tot}}_s(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})) \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$. Applying the same argument with $-s$ gives the reverse inclusion, hence equality.

Finally, if $\mathcal{N}$ is any von Neumann subalgebra with the same containment and stability properties, then $\mathcal{N} \in \mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$. Since $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$ is the intersection of all members of that class, it is contained in $\mathcal{N}$. This is exactly the desired minimality. □

Proposition 7.4 (Stable-hull transfer lies in a modular exact fixed sector)**.** *Assume:*

1. *the hypotheses of Proposition 5.3;*
2. *FSA and the exact pointer-invariance condition of Proposition 7.2;*
3. *MOD holds for* $\mathcal{A}_{\mathrm{fix}}$.

Then

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. By Proposition 7.2, $\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\mathrm{fix}}$. FSA makes $\mathcal{A}_{\mathrm{fix}}$ a von Neumann algebra, MOD makes it invariant under the ambient modular flow, and exact fixedness implies forward stability under the decoherence semigroup. Hence $\mathcal{A}_{\mathrm{fix}} \in \mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$. Since $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$ is the intersection of all members of that class, it is contained in $\mathcal{A}_{\mathrm{fix}}$. □

Corollary 7.5 (Partial Lemma C)**.** *Assume the hypotheses of Proposition 7.3. Then* $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$ *is a von Neumann subalgebra of* $\mathcal{A}_U$*, and the restricted vector state*

$$\omega^{\mathrm{st}}_{\mathfrak{F}} := \langle \Omega, \cdot\, \Omega \rangle|_{\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})}$$

is faithful. If, in addition, the hypotheses of Proposition 7.4 hold, then

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. The subalgebra statement is exactly Proposition 7.3. Because Ω is separating for the ambient algebra $\mathcal{A}_U$, it is also separating for every von Neumann subalgebra of $\mathcal{A}_U$; hence the restricted vector state is faithful on $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$. The inclusion claim is Proposition 7.4. □

Anchored shell existence around the stable hull. The stable-hull output already anchors a Foundation I shell problem. Write

$$\mathcal{A}_* := \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}).$$

For $X \in \mathrm{Ball}(\mathcal{A}_U)$ and $q \geq 0$, define the $\mathcal{A}_*$-relative defect

$$\delta^{\mathcal{A}_*,\Omega}_q(X) := \mathrm{dist}^{\sharp}_{\Omega}\big(\mathcal{E}_q(X),\, \mathrm{Ball}(\mathcal{A}_*)\big), \qquad \|Y\|^{\sharp}_{\Omega} := \big(\|Y\Omega\|^2 + \|Y^*\Omega\|^2\big)^{1/2}.$$

For $\varepsilon \geq 0$ and $\tau > 0$, let

$$\mathfrak{R}^{\mathcal{A}_*,\Omega}_{\varepsilon,\tau} := \left\{ \mathcal{M} \;\middle|\; \begin{array}{l} \mathcal{A}_* \subseteq \mathcal{M} \subseteq \mathcal{A}_U,\ \mathcal{M} \text{ is a unital vN subalgebra,} \\ \delta^{\mathcal{A}_*,\Omega}_q(X) \leq \varepsilon,\ \forall X \in \mathrm{Ball}(\mathcal{M}),\ \forall q \in \mathbb{Q} \cap [0,\tau] \end{array} \right\}.$$

Proposition 7.6 (Anchored relative-shell existence around the stable hull)**.** *Assume the hypotheses of Corollary 7.5. Then for every* $\varepsilon \geq 0$ *and* $\tau > 0$*:*

1. *the class $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$ is nonempty;*

2. *every chain in $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$ has an upper bound;*

3. *therefore maximal elements exist.*

Proof. By Corollary 7.5, $\mathcal{A}_*$ is a von Neumann subalgebra of $\mathcal{A}_U$ and is forward-stable under the decoherence semigroup. Hence $\mathcal{A}_*$ itself belongs to $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$: for $X \in \mathrm{Ball}(\mathcal{A}_*)$, forward stability gives $\mathcal{E}_q(X) \in \mathcal{A}_*$, and contractivity gives $\mathcal{E}_q(X) \in \mathrm{Ball}(\mathcal{A}_*)$, so $\delta_q^{\mathcal{A}_*,\Omega}(X) = 0$. Thus the class is nonempty.

Let $\{\mathcal{M}_i\}_{i\in I}$ be a chain in $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$, and set

$$\mathcal{M} := W^*\Big(\bigcup_{i\in I} \mathcal{M}_i\Big).$$

Then $\mathcal{A}_* \subseteq \mathcal{M} \subseteq \mathcal{A}_U$, and $\mathcal{M}$ is a unital von Neumann subalgebra. By Kaplansky density, every $X \in \mathrm{Ball}(\mathcal{M})$ is a bounded strong* limit of a net from the unit ball of the algebraic union $\bigcup_i \mathcal{M}_i$. Since the chain is totally ordered, each approximant lies in some admissible member of the chain, so its $\mathcal{A}_*$-relative defect is bounded by ε for all $q \in \mathbb{Q} \cap [0,\tau]$. Passing to the limit in the $\Omega^\sharp$-seminorm yields the same bound for X, hence $\mathcal{M} \in \mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$. Therefore every chain has an upper bound. Zorn's lemma now gives a maximal element. □

Remark 7.7 (Status of anchored shells). Proposition 7.6 does not yet construct the full shell family required on the Paper I side. It closes only the anchored existence problem around the theorem-level Foundation I output $\mathcal{A}_*$. What remains open is the upgrade to a canonical, nontrivial, and calibrated shell family with the correct modular transport law.

Remark 7.8 (Why equality is not the right first target). In general, $\mathcal{A}_{\mathrm{fix}}$ may be much larger than the minimal modular-decoherence stable hull of finitely many foreground pointer observables. The honest first target is therefore the inclusion

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}},$$

not the equality

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \stackrel{?}{=} \mathcal{A}_{\mathrm{fix}}.$$

The equality problem should remain open until one proves that the foreground realization together with the stable-hull transfer generates the full exact fixed sector. A further typed interface would still be needed to relate either object to probe-accessible effects or to the environmental record algebra.

Remark 7.9 (Relation to Paper III). One should not identify the foreground complexity k^* directly with the statistical complexity parameter n of Paper III [2]. The safer statement is only that $\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$ provides a natural algebraic candidate for a later bridge to the finite-observer sector studied there.

8 What Remains Open

The bottlenecks are now sharper than they were before this note.

1. **Persistent projection selection.** Lemma 3.2 is only conditional. The constructive question of which projection emerges from the source datum remains open.

2. **Foreground realization and model-independent PS2.** The stable-hull route closes the transfer-map well-definedness problem once $\mathcal{A}_{\mathfrak{F}}$ exists. Within the restricted product-classical and finite-dimensional faithful pointer regime, the PS2 package is now theorem-backed: Lemma 6.2 closes the structural reduction to fragment-local omission losses, Lemma 6.4 closes orthogonality, and Corollary 6.10 packages the restricted selection theorem. Lemma 6.12 now fixes the bookkeeping split of the survival deficit into a pointer-diagonal term and an off-sector remainder. In addition, Proposition 6.14 and Corollary 6.16 now show that the foreground-background part of the model-independent programme closes conditionally once four tail inputs are granted: approximate tail optimality, exchange-cover plus persistence, tail-passive background survival, and an exchange-efficiency gap, together with the block-locality defects for the KL and control terms. Corollary 6.17 then reduces the remaining diagonal product-defect problem to finitely many ordered within-block prefix couplings. The remaining open issue is therefore to derive those tail hypotheses together with the prefix-level coupling and locality bounds model-independently from the broken-phase output and the full T-DOME survival identity rather than assume them. Beyond the restricted regime, the unresolved analytic obstacles are explicit: cross-channel correlations need not decouple, pointer spectra need not remain finite or discrete, and the Hilbert–Schmidt control used here is not available in general Type III infinite-dimensional settings, where a predual reformulation may be necessary. The pointer-sector existence input PS1 is treated here only as an exact-regime idealization; what remains genuinely open is the model-independent alignment step PS2 and the induced ordered-selection step PS3.

3. **Comparison with typed HAFF objects.** The present paper proves inclusion only into the exact fixed sector under FSA and MOD. Equality with that sector remains open, and no canonical equivalence with probe-accessible effects or the environmental record algebra is asserted.

4. **Bridge to Paper I admissibility.** To feed directly into Paper I, one must still verify that the algebra produced by the lower bridge satisfies the additional admissibility hypotheses used there, beyond the already-closed faithfulness condition. More precisely, the remaining interface is to supply the independent dynamical modular-decoherence scaling axiom (C) together with an operational shell-generation/closure interface $\{\mathcal{A}_*(\tau)\}_{\tau>0}$ around the stable-hull output $\mathcal{A}_* := \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$, and one-scale cyclicity of the corresponding shell-annulus algebra
$$\mathcal{D}_\tau := \mathcal{A}_*(\tau) \cap \mathcal{A}_*'.$$
In Paper I language these are exactly the remaining inputs $(C)+(Amb)+(CN_\exists)$. Foundation I does not derive the shell-generation rule or its covariance. The latter is to be derived in Paper I from the independently stated dynamical axiom (C), once membership and closure are supplied. The shell-family side has only partially internalized: Proposition 7.6 closes anchored relative-shell existence around $\mathcal{A}_*$. What remains open is the canonical and calibrated nontrivial shell family. The naive scalar-threshold route now has an exact blocker: fixed thresholds are compatible with modular transport but not with exact core calibration, vanishing thresholds give exact core calibration but not automatic modular transport, and the only scalar profile with both properties is the trivial zero-threshold profile, which collapses the shells back to $\mathcal{A}_*$. More sharply, even the fixed-threshold branch now reduces to finding a scale-invariant nontriviality window between core collapse and ambient collapse, together with uniqueness of maximal shells inside that window; exact calibration is a separate issue beyond that.

These open problems are real, but they are no longer vague. Foundation I is now separated into one conditional source-selection problem, one theorem-level thermodynamic packaging result, one theorem-level stable-hull transfer construction, and one remaining foreground-realization/comparison problem.

9 Conclusion

The purpose of this paper has been calibration, not inflation. Foundation I is neither absent nor complete. It is partially available in theorem-level form, partially available as an explicit construction problem, and partially still conditional.

The main positive result is that the T-DOME I–II chain can already be written as a formal Foundation I proposition, and that the first restricted PS2 bottleneck can now be closed at theorem level in a first restricted regime. In the product-classical, decoherence-dominated pointer regime, pointer coarse-graining yields fragment-local omission losses, independent fragment records force orthogonality, semigroup decay plus explicit Hilbert–Schmidt continuity constants bound the coherence remainder, and a sufficiently large weight gap forces exact top-k^* pointer selection. On the algebraic side, the first operator-algebraic realization of the foreground sector can now be extended to a well-defined minimal modular-decoherence stable hull, yielding a faithful von Neumann subalgebra together with the inclusion

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\mathrm{HAFF}}.$$

At the level of future PS2 work, the paper now also fixes the bookkeeping split of the survival deficit into a pointer-diagonal term and an off-sector remainder, and it closes the first foreground-background decoupling step conditionally: if the tail distortion localizes, the exchange-weighted defect persists, the background diagonal survival becomes tail-passive, the exchange efficiency ratio stays below the squared-loss threshold, and the remaining KL/control locality defects are small, then the broken phase forces tail-small diagonal foreground-background coupling and hence tail-small foreground-background mutual information. Corollary 6.17 then shows that the unresolved product-defect part is no longer a monolithic statement: it is a finite family of ordered within-block prefix decoupling problems. The remaining model-independent problem is therefore no longer vague: it is to derive those tail hypotheses and the ordered prefix-level coupling/locality bounds, and thereby to control the full diagonal product defect as a theorem-level output of the broken phase. On the Paper I side, the admissibility interface is now sharpened in the same way: faithfulness is already closed, ambient scaling covariance would transfer the dynamical axiom (C), and the remaining regularity burden is exactly the existence of a Foundation I shell family and one cyclic shell-annulus scale. Even the shell-family side is no longer a black box: anchored relative shells around $\mathcal{A}_*$ exist, but their upgrade to a canonical nontrivial shell family currently stops at the scalar-threshold trilemma between transport, exact core calibration, and nontriviality. Within the fixed-threshold route, this has already reduced further to a scale-invariant nontriviality window and uniqueness problem, plus a separate calibration theorem. The main negative result is equally important: nothing in the present note licenses the stronger claim that the source datum has already been shown to generate the full accessible algebra uniquely, nor that PS2 has been derived model-independently from the broken-phase output. Those remain the unfinished parts of the unified-source programme.

In that sense, the current status is simple. The thread can be strung, but only honestly: conditional at the source end, inclusion-first at the algebra end, and theorem-backed only where the intermediate steps have actually been closed.

A External Genesis Inputs (Self-Contained Statements)

The following list records the external Genesis-side inputs used in the paper in a self-contained form. The point is not to reproduce their full proofs, but to state exactly what is imported from HAFF and T-DOME.

1. **HAFF ambient datum.** The ambient setting is a Type III_1 von Neumann algebra in standard form $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, with Ω cyclic and separating for $\mathcal{A}_U$, together with a pointwise σ-weakly continuous semigroup of normal completely positive unital maps $\{\mathcal{E}_t\}_{t\geq 0}$ on $\mathcal{A}_U$. This is the source datum formalized in Section 2.

2. **T-DOME I projected record dynamics.** Given a persistent Nakajima–Zwanzig projection $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$, the projected predual dynamics is analyzed in a non-Markovian kernel form

$$\partial_t \omega_{\mathcal{P},t} = \int_0^t \mathcal{K}_*(t,s)\, \omega_{\mathcal{P},s}\, ds \; + \; \text{inhomogeneous term},$$

   or equivalently in the dual memory-kernel formulation used in Section 2. Foundation I uses only the existence of such a kernel and a persistent record process, not an explicit closed-form solution.

3. **T-DOME I survival identity and distortion functional.** The survival bookkeeping imported into the PS2 analysis is the identity

$$\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}} - \beta\, \Delta\langle H_{\mathrm{ctrl}}\rangle,$$

   with frame comparisons measured in the squared-loss distortion

$$D(\mathcal{F}_I) := \mathbb{E}_\xi\Big[\big(\mathcal{S}_{\mathrm{full}}(\xi) - \mathcal{S}_{\mathcal{F}_I}(\xi)\big)^2\Big].$$

   The diagonal/off-sector split and all restricted PS2 estimates are built from this identity rather than assumed independently.

4. **Finite-coordinate compression model motivated by T-DOME II.** T-DOME II motivates a coordinate-coding problem, but its universal computational-ceiling and SSB-necessity claims are not imported. Proposition 4.1 instead assumes a rate charge, strict improvement up to the feasible rank, and an irreducible group action. Only in that supplied FD/I model does $k^* = \lfloor \dot{\mathcal{C}}_{\mathrm{budget}}/h_\mu \rfloor$ and the selection of a proper foreground subspace follow.

5. **Decoherence-side pointer sector (exact-regime idealization).** In the exact dephasing-dominated regime, robust classical records are carried by a commuting pointer sector. Foundation I uses this as the PS1 input: a commuting family of exact fixed-point projections for $\{\mathcal{E}_t\}$ generating a classical pointer algebra inside $\mathcal{A}_U$. This is an idealization of the approximate einselection and redundant-record structure described in the quantum Darwinism literature [9, 10], not a consequence of generic decoherence alone.

6. **Exact fixed sector, not a universal accessibility object.** The inclusion comparison uses $\mathcal{A}_{\mathrm{fix}} = \mathrm{Fix}(\mathcal{E})$ only under FSA and MOD. This object is an exact noiseless/fixed sector. It is not identified with the probe-accessible operator system or the environmental pointer/record algebra.

7. **T-DOME III / Paper III calibration distinction.** Foundation I uses T-DOME III and Paper III only as contextual support for what is *not* claimed: the foreground complexity k^* is not identified with Paper III's statistical complexity parameter n, and the present paper does not attempt a direct derivation of Paper III's self-description bounds from the source-side chain.

B Conditional Input Ledger

For ease of reference, the remaining conditional interfaces are listed here in one place.

1. **Source selection.** Lemma 3.2 is conditional on the existence of a persistent Nakajima–Zwanzig projection producing the projected record kernel. This is the observer-emergence bottleneck. No general semigroup-gap, CP-divisibility, or kernel-decay criterion is asserted here: each would require a specified persistence functional and can fail to capture the non-Markovian record regime. A rigorous sufficient criterion remains open rather than being installed for downstream use without proof.

2. **Pointer alignment package.** The foreground realization problem separates into:

   (PS1) existence of a decoherence-selected exact fixed-point pointer sector (exact dephasing-dominated regime idealization, not a consequence of generic decoherence);

   (PS2) identification of the T-DOME sufficient-statistic coordinates with the pointer-sector coordinates;

   (PS3) ordered pointer-channel selection by survival relevance.

   PS1 is an exact-regime idealization of the decoherence-side input; the genuine model-independent open problem is PS2, with PS3 downstream from it.

3. **Exact fixed-sector structural inputs.** The main text keeps the following conditions separate because none is a consequence of the bare Type III source datum:

   (FSA) $\mathrm{Fix}(\mathcal{E})$ is closed under the ambient product and is a von Neumann subalgebra;

   (MOD) that fixed algebra is invariant under the ambient modular flow;

   (EXP) when an expected subalgebra is required, an ω-preserving faithful normal conditional expectation exists;

   (PTR) the separately specified environmental pointer algebra is contained in the exact fixed algebra in the idealized noiseless regime.

   Faithfulness of ω alone does not establish MOD. In the faithful standard-form regime, Takesaki's theorem relates MOD to an ω-preserving expectation only after FSA supplies the subalgebra to which the theorem applies [12, 13].

4. **Restricted PS2 regime assumptions.** The theorem-level restricted PS2 closure is proved only in the product-classical, finite-dimensional faithful pointer regime stated in Section 6. In particular, the restricted closure does not yet treat general cross-channel correlations, continuous pointer spectra, or the genuinely infinite-dimensional Type III estimate problem.

5. **Conditional foreground-background decoupling inputs.** Proposition 6.14 requires the following named inputs on the tail regime: approximate tail optimality, exchange-cover plus persistence, tail-passive background survival, and an exchange-efficiency gap, together with the KL/control locality defects used in Corollary 6.16.

6. **Abstract compatibility of the four tail packages.** An engineered finite classical Markov witness realizes the four tail packages simultaneously. This is an FD/I audit witness establishing joint mathematical compatibility, not a physical derivation and not a Type III shell model.

7. **Natural physical nonvacuity remains open.** No natural source/control model is known to derive all four packages jointly, and no known model simultaneously realizes those packages and the modular shell interface. Proposition 6.14 therefore remains conditional despite the compatibility witness.

8. **Conditional full product-defect inputs.** Corollary 6.17 adds the ordered within-block prefix-chain analogues of the same decoupling/locality inputs. The full diagonal product-defect problem is therefore reduced to finitely many named prefix couplings rather than left as one undifferentiated gap.

9. **Paper I admissibility inputs.** The lower output feeds into Paper I only conditionally on three further interfaces: the independent dynamical covariance axiom (C), an operational shell membership/closure rule around the stable hull, and one-scale shell-annulus cyclicity. Proposition 7.6 closes only an anchored existence problem. It does not derive the Paper I shell interface. In particular, covariance is not placed into that interface; Paper I must derive shell scaling from (C).

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Foundation II: Shared Manifestation
A Conditional Two-Source Program for Shared Accessible Sectors

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Abstract

Foundation I of the Consciousness Series [1] constructs a conditional chain from a single source datum $S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\})$ to an admissible accessible pair $(\mathcal{A}_c, \omega)$, which then feeds into the inseparability theorem of Paper I [2]. That chain is single-source: it produces one observer's manifest sector. The present paper, Foundation II, asks the next structural question: when two source chains share the same ambient substrate, under what conditions does a non-trivial *shared* manifest sector emerge?

We define a multi-source datum as a family of source data sharing a common ambient triple $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, each equipped with its own decoherence semigroup. No global tensor-product decomposition is assumed; the multi-source structure is encoded entirely through subalgebra inclusions, following the non-canonicality of tensor-product factorizations in the Type III_1 setting established in [5]. We then introduce joint-stability conditions under which a candidate shared sector $\mathcal{A}_{\text{sh}}$ is simultaneously stable for both sources.

The paper's main result is a conditional two-source foothold, not a general multi-source closure theorem. Under named joint-stability and compatible-projection conditions, we construct a shared stable hull $\Phi^{\text{st}}_{\text{sh}}(\mathfrak{F}_{\text{sh}})$ from the overlap of the two foreground frames, prove that it is a von Neumann subalgebra carrying a faithful restricted state, and show that, under an additional admissibility input, the resulting pair $(\Phi^{\text{st}}_{\text{sh}}(\mathfrak{F}_{\text{sh}}), \omega^{\text{st}}_{\text{sh}})$ lies in the theorem domain of Paper I. The shared sector therefore inherits the geometry–modular-flow inseparability of the single-source case.

After the main theorem, we analyze a regime-dependent common/private structure: under a shared/private regime assumption, the shared sector is the locus on the overlap algebra where the two manifest states agree, while observer-dependent discrepancy can only appear outside that shared sector. Finally, we state a conditional N-source extension under pairwise-threshold and exchangeability assumptions, and record the open problems that remain, including the quantitative relationship between consensus strength and geometric stability.

This paper treats only the mathematical bridge from multi-source data to shared accessible sectors. It does not claim a general N-source theorem, does not provide a quantitative consensus–stability bound, and does not derive the candidate shared sector from bare multi-source dynamics alone. It also does not attempt to settle questions of intersubjective ontology.

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1 Introduction

Foundation I of the Consciousness Series [1] addressed the single-source problem. Starting from a source datum

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

it isolated a conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where $\mathcal{O}_A$ is a persistent projected record process, $\mathcal{O}_B$ is a foreground frame supplied under explicit finite-coordinate coding assumptions, and $(\mathcal{A}_c, \omega)$ is the manifest accessible pair that then feeds into the geometry–modular-flow inseparability theorem of Paper I [2]. That was the correct scope for the first foundation paper: it explained how one observer-facing manifest sector could arise from one source chain.

The present paper asks the next structural question. If two source chains share the same ambient substrate, under what conditions does a non-trivial *shared* manifest sector arise? This is the minimum multi-source problem behind any later discussion of a shared world. The paper does not attempt to settle the full intersubjective problem in one step. It does not claim a general N-source theorem, does not derive a quantitative consensus-strength/stability law, and does not aim to settle questions of ontology. Its narrower aim is to identify a theorem-level foothold for *shared manifestation existence*.

That calibration matters. In the current programme there are two separate bridges beyond the already-written Consciousness Trilogy results. The first concerns whether entangled or correlated source chains can support a common manifest sector at all. The second concerns how the strength of that commonality controls the quantitative stability of the resulting geometry. Foundation II addresses only the first. The second remains a separate Bridge 3 problem and should not be smuggled into the present paper under vague language about "consensus" or "robustness".

The paper therefore adopts the smallest honest target. Its main theorem is proved in the two-source case. The theorem does not directly state that a full shared geometry has been derived. Its first conclusion is weaker and mathematically safer: under named compatibility and joint-stability inputs, one can construct a *shared admissible manifest sector*

$$(\mathcal{A}_{\text{sh}}, \omega_{\text{sh}}) \subseteq (\mathcal{A}_c^{(1)}, \omega_1) \cap (\mathcal{A}_c^{(2)}, \omega_2),$$

which is stable under the relevant source-side dynamics and can then be fed into the already-existing Paper I machinery. Geometry enters only as a corollary once that admissible shared sector is in hand. The main result should therefore be read as a conditional inclusion theorem and theorem-level foothold, not as a full emergence theorem for shared worlds.

At the heuristic level, this shared-sector question is related to the quantum Darwinism and spectrum-broadcast-structure literature, where the objective part of a quantum description is identified with redundantly accessible pointer information carried across multiple environment fragments [17, 18, 19, 20]. The present paper does not translate those frameworks into a theorem about Type III_1 ambient algebras. It only records the operator-algebraic analogue needed here: a shared foreground overlap together with a shared stable hull.

Three design choices govern the whole paper.

1. **Two sources first.** The paper starts at $N = 2$, because the two-source case is the minimum non-trivial setting in which one can ask whether a shared manifest sector exists at all. General N-source statements are postponed to a conditional extension section.

2. **No tensor-product starting point.** The multi-source datum is formulated inside a common ambient algebra rather than from a canonical tensor decomposition. This follows the HAFF viewpoint that physically relevant subsystem structure is selected by stability and accessibility rather than fixed in advance [5, 6].

3. **Common/private decomposition is secondary.** After the main theorem, we analyze how each observer's full manifest algebra may be written, under a regime assumption, as a shared sector plus a private complement. That decomposition is treated as a structural consequence, not as a competing main theorem.

The paper's mathematical contributions are correspondingly limited and explicit.

1. We define a *multi-source datum* on a common standard-form von Neumann algebra, with Type III_1 as the primary regime, together with independent faithful normal source states and source-specific decoherence semigroups.

2. We separate the new inputs into three named joint-stability conditions: shared-sector invariance, shared-state existence with faithfulness, and projection/kernel compatibility.

3. In the two-source case, we define the shared foreground overlap and the associated shared stable hull, and prove the abstract minimality properties of that hull.

4. Under an explicit candidate shared-sector input, we prove an inclusion theorem showing that the shared stable hull lies inside the common manifest carrier, and therefore yields a shared admissible manifest sector.

5. Under a shared/private regime assumption, we analyze the resulting common/private structure by identifying the shared sector as the locus of agreement on the overlap algebra and by localizing overlap-level discrepancy outside that sector.

The role of conditional language is exactly the same as in Foundation I. Wherever a piece can be internalized cleanly, we internalize it. Wherever it cannot, we isolate it and keep it explicit. The present theorem does not derive the existence of a candidate shared sector from bare multi-source dynamics or entanglement alone. It starts from such a sector as named input and analyzes the stable closure that follows from it. The main open points here are not hidden: model-independent joint stability, compatible Nakajima–Zwanzig projections, and the well-definedness of common/private decomposition in the Type III_1 setting.

The paper is organized as follows. Section 2 defines the multi-source datum and the joint-stability package. Section 3 states and proves the two-source main theorem. Section 4 analyzes the common/private structure under a regime assumption. Section 5 gives a conditional N-source extension. Section 6 records the open problems. Section 7 collects the named conditional inputs. Appendix A summarizes the imported Genesis and Consciousness results used by the paper.

2 Multi-Source Datum and Joint-Stability Inputs

The first task is to define the source-side object language without overloading it with later conditional interface data.

Definition 2.1 (Multi-source datum)**.** A *multi-source datum* is a tuple

$$S_{\text{multi}} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_i\}_{i=1}^N, \{\mathcal{E}_t^{(i)}\}_{i=1}^N),$$

where:

1. $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a von Neumann algebra in standard form, with Ω cyclic and separating for $\mathcal{A}_U$; the programme's primary infinite-dimensional regime is a Type III_1 factor, while explicitly marked model checks may be finite Type I;

2. $\varpi_1, \ldots, \varpi_N$ are independently specified faithful normal states on $\mathcal{A}_U$; no agreement of their restrictions to later overlap algebras is part of the datum;

3. for each $i \in \{1, \ldots, N\}$, $\{\mathcal{E}_t^{(i)}\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$.

Remark 2.2 (Heisenberg-picture convention). The semigroups $\{\mathcal{E}_t^{(i)}\}$ act on observables, so the paper uses the Heisenberg-picture convention throughout. The continuity assumption in Definition 2.1 is therefore pointwise σ-weak continuity on $\mathcal{A}_U$, not norm continuity.

Remark 2.3 (A deliberately thin datum). The multi-source datum contains only the common ambient standard-form triple, the independent source states, and the family of source-side dynamical semigroups. It does *not* include Nakajima–Zwanzig projections, foreground frames, accessible algebras, or any private/shared decomposition. Those are not primitive data; they are interface or output objects that may or may not be obtainable from the datum under additional assumptions.

Remark 2.4 (Why no tensor-product factorization). The present paper does not assume a global tensor-product decomposition

$$\mathcal{H}_U \cong \bigotimes_{i=1}^N \mathcal{H}_i \qquad \text{or} \qquad \mathcal{A}_U \cong \bar{\otimes}_{i=1}^N \mathcal{A}_i.$$

This is deliberate. In the HAFF programme, the physically relevant subalgebras are selected by accessibility and stability rather than by a canonical prior factorization [5, 6]. In the Type III setting, treating the subsystem split as primitive would import exactly the kind of structure the Genesis programme is designed to derive rather than presuppose.

The single-source outputs from Foundation I are now imported only as downstream conditional objects.

Definition 2.5 (Individual manifest outputs)**.** Let S_{multi} be a multi-source datum. For each source index i, an *individual manifest output* means a pair

$$(\mathcal{A}_c^{(i)}, \omega_i)$$

obtained from the Foundation I chain for the i-th source, where $\mathcal{A}_c^{(i)} \subseteq \mathcal{A}_U$ is a von Neumann subalgebra and

$$\omega_i := \varpi_i|_{\mathcal{A}_c^{(i)}}$$

is its faithful normal manifest state.

Remark 2.6 (Independent states are load-bearing). The states ω_i are not all restrictions of the reference vector state $\langle \Omega, \cdot\, \Omega \rangle$. The vector Ω fixes a standard-form representation and the reference modular flow used below; the ϖ_i carry source-specific state data. If instead one defined every ω_i by restricting a single ambient state, then the states would agree automatically on every intersection and the overlap discrepancy of Section 4 would vanish identically.

Remark 2.7. Nothing in Definition 2.3 claims that every source in a multi-source datum automatically produces an individual manifest output. That is already the conditional content of Foundation I [1]. Foundation II takes those single-source outputs as antecedent material when the two-source main theorem is stated.

Definition 2.8 (Candidate shared sector)**.** Given individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$, a *candidate shared sector* is a von Neumann subalgebra

$$\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \bigcap_{i=1}^{N} \mathcal{A}_c^{(i)}.$$

The new source-side content begins with the joint-stability package. We state it in three layers so that later failures can be localized cleanly.

Definition 2.9 (JS-A: shared-sector invariance and semigroup compatibility)**.** Let S_{multi} be a multi-source datum and let $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{A}_U$ be a candidate shared sector. We say that $(\mathcal{A}_{\text{sh}}^{\text{cand}}, \{\mathcal{E}_t^{(i)}\})$ satisfies *JS-A* if:

1. $\mathcal{E}_t^{(i)}(A) \in \mathcal{A}_{\text{sh}}^{\text{cand}}$ for every $A \in \mathcal{A}_{\text{sh}}^{\text{cand}}$, every i, and every $t \geq 0$;
2. the ambient modular flow $\sigma_s^{\omega_0,\text{tot}}$ of the reference state $\omega_0(\cdot) = \langle \Omega, \cdot\, \Omega \rangle$ satisfies
$$\sigma_s^{\omega_0,\text{tot}}(\mathcal{A}_{\text{sh}}^{\text{cand}}) = \mathcal{A}_{\text{sh}}^{\text{cand}} \qquad \forall s \in \mathbb{R};$$
3. the restricted semigroups commute on the candidate shared sector:
$$\mathcal{E}_t^{(i)} \mathcal{E}_u^{(j)}(A) = \mathcal{E}_u^{(j)} \mathcal{E}_t^{(i)}(A) \qquad \forall A \in \mathcal{A}_{\text{sh}}^{\text{cand}},\ \forall t, u \geq 0.$$

Remark 2.10 (Status of the JS-A commutativity clause). The exact commutativity requirement in Definition 2.9 is a strong regime assumption. The present paper does not claim that it follows model-independently from the bare multi-source datum. A weaker variant would replace strict commutativity by asymptotic commutativity in the long-time limit, or by commutativity only on the shared fixed-point algebra. The stronger form is used here because it makes the shared stable-carrier construction transparent; the weaker variants are left open.

Definition 2.11 (JS-B: shared-state existence and faithfulness)**.** Let $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{A}_U$ be a candidate shared sector. We say that $\mathcal{A}_{\text{sh}}^{\text{cand}}$ satisfies *JS-B* if there exists a faithful normal state

$$\omega_{\text{sh}} \in (\mathcal{A}_{\text{sh}}^{\text{cand}})_*$$

such that:

1. $\omega_{\text{sh}} \circ \mathcal{E}_t^{(i)} = \omega_{\text{sh}}$ on $\mathcal{A}_{\text{sh}}^{\text{cand}}$ for every i and every $t \geq 0$;
2. ω_{sh} is compatible with the individual manifest states in the sense that
$$\omega_i|_{\mathcal{A}_{\text{sh}}^{\text{cand}}} = \omega_{\text{sh}} \qquad \forall i \in \{1, \ldots, N\}.$$

Remark 2.12 (JS-B is an agreement assumption, not a discovery). Faithfulness of each restriction follows from faithfulness of the independent ambient state ϖ_i, but equality of two restrictions does not. Clause (2) of JS-B is therefore the explicit substantive agreement input. If all ω_i were restrictions of the single reference state ω_0, clause (2) would be automatic and the discrepancy analysis would have no nonzero cases. The present datum deliberately excludes that shortcut.

Remark 2.13 (Strength of the JS-B stationarity clause). The stationarity requirement in Definition 2.11 is strong. It amounts to asking that the shared state lie in the common fixed-state sector of all source semigroups when restricted to $\mathcal{A}_{\text{sh}}^{\text{cand}}$. The present paper does not claim that such a common stationary state is generic. A weaker variant would replace exact stationarity by long-time convergence or approximate stationarity on the shared sector; those relaxations are left open.

Definition 2.14 (JS-C: compatible persistent projections)**.** Let S_{multi} be a multi-source datum, and fix a candidate shared sector $\mathcal{A}_{\text{sh}}^{\text{cand}}$. A pair of Nakajima–Zwanzig projections

$$\mathcal{P}^{(1)}, \mathcal{P}^{(2)} : \mathcal{A}_U \to \mathcal{A}_U$$

is called $\mathcal{A}_{\text{sh}}^{\text{cand}}$-*compatible* if:

1. each $\mathcal{P}^{(i)}$ is a normal idempotent projection used in the Stage A output of the i-th source chain;
2. they commute,
$$\mathcal{P}^{(1)}\mathcal{P}^{(2)} = \mathcal{P}^{(2)}\mathcal{P}^{(1)} =: \mathcal{P}_{\text{sh}};$$
3. $\mathcal{P}_{\text{sh}}(\mathcal{A}_U)$ is a von Neumann subalgebra containing $\mathcal{A}_{\text{sh}}^{\text{cand}}$, and $\mathcal{P}_{\text{sh}}|_{\mathcal{A}_{\text{sh}}^{\text{cand}}} = \text{id}$;
4. the projected predual dynamics induced by the pair admit a common dual memory kernel
$$\mathcal{K}_{\text{sh},*}(t, s)$$
on the shared projected sector, meaning that there exists a single kernel governing the shared projected dynamics on that sector; the definition does not require literal equality of the two individual source kernels before projection to the shared sector.

Remark 2.15 (Status of JS-C). Definition 2.14 is not imported from a standard open-systems textbook. It is a new named compatibility condition introduced for the present paper. The point is not to pretend that a model-independent notion of compatible Nakajima–Zwanzig projections is already standard. The point is to isolate exactly what later proofs need from such a notion. The background projection formalism is still the standard Nakajima–Zwanzig one [11, 12]; what is new here is only the multi-source compatibility layer. A weaker variant would require only that the long-time asymptotic behaviour of the two projected kernels agree on the shared sector, rather than demanding a common kernel at all times. The present paper uses the stronger form because it simplifies the Stage A construction; the weaker variant is left as an open relaxation.

Proposition 2.16 (Commuting expectations supply the algebraic part of JS-C)**.** *Let $E_i : \mathcal{A}_U \to \mathcal{M}_i$, $i = 1, 2$, be normal conditional expectations. If $E_1E_2 = E_2E_1$, then*

$$E_{12} := E_1E_2$$

is a normal conditional expectation onto $\mathcal{M}_1 \cap \mathcal{M}_2$. Consequently, if $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{M}_1 \cap \mathcal{M}_2$, clauses (1)–(3) of JS-C hold with $\mathcal{P}^{(i)} = E_i$. Clause (4), existence of a common projected memory kernel, remains an additional dynamical input.

Proof. Commutation and idempotence give $E_{12}^2 = E_1^2 E_2^2 = E_{12}$. Its range is contained in both $\mathcal{M}_1$ and $\mathcal{M}_2$, while every element of the intersection is fixed by both expectations. Thus $\operatorname{Ran} E_{12} = \mathcal{M}_1 \cap \mathcal{M}_2$. Normality, complete positivity, and unitality pass to the composition; the resulting norm-one positive projection is a conditional expectation. This is the commuting-expectation/commuting-square mechanism, not an automatic property of arbitrary Nakajima–Zwanzig projections [14, 15]. □

Proposition 2.17 (Finite-dimensional two-source witness)**.** *There is a finite Type I model with independent source states satisfying JS-A, JS-B, and JS-C and with $k_{\mathrm{sh}} = 1$. In the same model, the two manifest states agree on the shared sector but have nonzero discrepancy elsewhere in their overlap.*

Proof. Let

$$\mathcal{A}_U = M_2(S) \otimes M_2(P_1) \otimes M_2(P_2)$$

in its tracial standard form, and let Z_X be Pauli Z on factor X. Write

$$D_X(A) := \tfrac{1}{2}(A + Z_X A Z_X)$$

for the dephasing expectation on factor X, tensored with the identity on the other factors. Set

$$\mathcal{P}^{(1)} = D_S D_{P_1}, \qquad \mathcal{P}^{(2)} = D_S D_{P_2}.$$

These normal UCP conditional expectations commute, and $\mathcal{P}_{\mathrm{sh}} = D_S D_{P_1} D_{P_2}$ has the full diagonal algebra as its range. For $\lambda_i > 0$, define

$$\mathcal{E}_t^{(i)} = e^{-\lambda_i t}\,\mathrm{id} + (1 - e^{-\lambda_i t})\mathcal{P}^{(i)}.$$

Idempotence of $\mathcal{P}^{(i)}$ gives the semigroup law; each map is normal, UCP, and the two semigroups commute.

Take

$$\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}} = W^*(Z_S) \otimes \mathbf{1} \otimes \mathbf{1}, \qquad \mathcal{A}_c^{(1)} = \mathcal{A}_c^{(2)} = W^*(Z_S, Z_{P_1}, Z_{P_2}).$$

The reference tracial modular flow is trivial, and both semigroups fix the candidate shared sector pointwise, proving JS-A. Let $\rho(x) := \frac{1}{2}(\mathbf{1} + xZ)$, with $|x| < 1$, and choose the independent faithful ambient states

$$\varpi_1 = \rho(1/3) \otimes \rho(1/4) \otimes \rho(1/5), \qquad \varpi_2 = \rho(1/3) \otimes \rho(-1/4) \otimes \rho(-1/5).$$

Their manifest restrictions agree faithfully on the shared sector, with common state induced by $\rho(1/3)$, and that state is stationary there. This proves JS-B. Proposition 2.16 proves the algebraic clauses of JS-C. On the pointwise-fixed shared sector the projected generator vanishes, so the common projected kernel may be taken as $\mathcal{K}_{\mathrm{sh},*} = 0$, proving clause (4).

Finally choose foreground images

$$V_{\mathrm{fg}}^{(1)} = \operatorname{span}\{Z_S, Z_{P_1}\}, \qquad V_{\mathrm{fg}}^{(2)} = \operatorname{span}\{Z_S, Z_{P_2}\}.$$

Their intersection is $\operatorname{span}\{Z_S\}$, hence $k_{\mathrm{sh}} = 1$. Yet $\omega_1(Z_{P_1}) = 1/4$ and $\omega_2(Z_{P_1}) = -1/4$, so the discrepancy is nonzero on the larger overlap algebra. This witness verifies compatibility in FD/I; its zero shared kernel and tracial modular flow are not a Type III realization of the full Foundation programme. □

3 The Two-Source Shared Manifest Sector

We now specialize to $N = 2$. The paper's main theorem is entirely two-source.

3.1 Joint source interface

Let

$$S^{(2)}_{\text{multi}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_1, \varpi_2\}, \{\mathcal{E}^{(1)}_t, \mathcal{E}^{(2)}_t\})$$

be a two-source datum. Assume that each source separately satisfies the named Foundation I hypotheses required to produce a Stage A output, Stage B output, and manifest pair

$$(\mathcal{A}^{(i)}_c, \omega_i), \qquad i = 1, 2.$$

Lemma 3.1 (Joint Stage A output from compatible persistent projections)**.** *Let $S^{(2)}_{\text{multi}}$ be a two-source datum. Assume there exists a candidate shared sector $\mathcal{A}^{\text{cand}}_{\text{sh}} \subseteq \mathcal{A}^{(1)}_c \cap \mathcal{A}^{(2)}_c$ satisfying JS-C. Then the commuting product*

$$\mathcal{P}_{\text{sh}} := \mathcal{P}^{(1)}\mathcal{P}^{(2)}$$

defines a shared projected record object

$$\mathcal{O}^{\text{sh}}_A := (\mathcal{P}_{\text{sh}}, \omega_{\text{sh},t}, \mathcal{K}_{\text{sh},*}(t, s)),$$

where $\omega_{\text{sh},t}$ is the projected normal state on the shared sector and $\mathcal{K}_{\text{sh},*}(t, s)$ is the common dual memory kernel from Definition 2.14.

Proof. Commutativity of $\mathcal{P}^{(1)}$ and $\mathcal{P}^{(2)}$ implies that $\mathcal{P}_{\text{sh}}$ is again an idempotent projection. Because both $\mathcal{P}^{(1)}$ and $\mathcal{P}^{(2)}$ are normal and they commute, their composition is again normal. By JS-C, $\mathcal{P}_{\text{sh}}(\mathcal{A}_U)$ is a von Neumann subalgebra containing $\mathcal{A}^{\text{cand}}_{\text{sh}}$, and the common dual kernel $\mathcal{K}_{\text{sh},*}(t, s)$ is part of the named input. This is exactly the shared Stage A object required for later stages. □

3.2 Joint foreground overlap

For each source $i = 1, 2$, let $\mathfrak{F}^{(i)}$ denote the foreground frame produced by the conditional Foundation I finite-coordinate compression interface, with foreground realization

$$\iota_i : V_{\text{fg}}(\mathfrak{F}^{(i)}) \hookrightarrow \mathcal{A}^{\text{sa}}_U.$$

Definition 3.2 (Shared foreground overlap)**.** The *shared foreground overlap* is the operator-side intersection

$$V_{\text{fg}}(\mathfrak{F}_{\text{sh}}) := \iota_1\big(V_{\text{fg}}(\mathfrak{F}^{(1)})\big) \cap \iota_2\big(V_{\text{fg}}(\mathfrak{F}^{(2)})\big) \subseteq \mathcal{A}^{\text{sa}}_U.$$

Whenever this intersection is non-zero and finite-dimensional, we choose a basis

$$O^{\text{sh}}_1, \ldots, O^{\text{sh}}_{k_{\text{sh}}} \in V_{\text{fg}}(\mathfrak{F}_{\text{sh}})$$

and write

$$\mathcal{A}_{\mathfrak{F}_{\text{sh}}} := W^*(1, O^{\text{sh}}_1, \ldots, O^{\text{sh}}_{k_{\text{sh}}}).$$

Proposition 3.3 (Dimension bound for the shared foreground)**.** *Assume $\mathfrak{F}^{(1)}$ and $\mathfrak{F}^{(2)}$ have effective dimensions k_1^* and k_2^*, respectively, and that the shared foreground overlap of Definition 3.2 is finite dimensional. Then*

$$k_{\text{sh}} \leq \min\{k_1^*, k_2^*\}.$$

Proof. The shared foreground overlap is a subspace of each individual foreground image. Therefore its dimension cannot exceed the dimension of either image, hence cannot exceed the minimum of the two. □

Remark 3.4. Proposition 3.3 is only a bookkeeping result. It does not prove that the overlap is non-trivial. Non-triviality of the shared foreground remains a separate conditional input.

3.3 Shared stable hull

Write

$$\omega_0(\cdot) = \langle \Omega, \cdot\, \Omega \rangle$$

on the ambient algebra $\mathcal{A}_U$, and let $\sigma_s^{\omega_0,\text{tot}}$ be its ambient modular flow.

Definition 3.5 (Shared stable-carrier class)**.** Let $\mathfrak{F}_{\text{sh}}$ be a non-trivial shared foreground overlap. Define

$$\mathfrak{L}_{\text{sh}}^{\text{st}} := \left\{ \begin{array}{l} \mathcal{M} \subseteq \mathcal{A}_U : \mathcal{M} \text{ is a von Neumann subalgebra,} \\ \qquad \mathcal{A}_{\mathfrak{F}_{\text{sh}}} \subseteq \mathcal{M}, \\ \qquad \sigma_s^{\omega_0,\text{tot}}(\mathcal{M}) = \mathcal{M} \quad \forall s \in \mathbb{R}, \\ \qquad \mathcal{E}_t^{(i)}(\mathcal{M}) \subseteq \mathcal{M} \quad \forall t \geq 0,\ i = 1, 2 \end{array} \right\}.$$

Definition 3.6 (Shared stable hull)**.** Assume $\mathfrak{L}_{\text{sh}}^{\text{st}} \neq \varnothing$. The *shared stable hull* of the shared foreground is

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) := \bigcap_{\mathcal{M} \in \mathfrak{L}_{\text{sh}}^{\text{st}}} \mathcal{M}.$$

Proposition 3.7 (Minimal shared stable hull)**.** *Assume $\mathfrak{L}_{\text{sh}}^{\text{st}} \neq \varnothing$. Then*

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})$$

is a well-defined von Neumann subalgebra of $\mathcal{A}_U$ satisfying:

1. $\mathcal{A}_{\mathfrak{F}_{\text{sh}}} \subseteq \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})$;
2. $$\sigma_s^{\omega_0,\text{tot}}\big(\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})\big) = \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \qquad \forall s \in \mathbb{R};$$
3. $$\mathcal{E}_t^{(i)}\big(\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})\big) \subseteq \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \qquad \forall t \geq 0,\ i = 1, 2;$$
4. *it is minimal with these properties: if $\mathcal{N} \subseteq \mathcal{A}_U$ is any von Neumann subalgebra containing $\mathcal{A}_{\mathfrak{F}_{\text{sh}}}$ and stable under the ambient modular flow and both semigroups, then*

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \subseteq \mathcal{N}.$$

Proof. The argument is identical to the stable-hull construction in Foundation I [1], with the only change that one now imposes stability under both source semigroups. An arbitrary intersection of von Neumann subalgebras is again a von Neumann subalgebra: intersections of σ-weakly closed $*$-subalgebras remain σ-weakly closed $*$-subalgebras. Because every member of $\mathfrak{L}^{\mathrm{st}}_{\mathrm{sh}}$ contains $\mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}}$, so does the intersection. Stability under the ambient modular flow and both semigroups is preserved by intersection: if $A \in \bigcap_\alpha \mathcal{M}_\alpha$, then $\mathcal{E}^{(i)}_t(A) \in \mathcal{M}_\alpha$ for every α, t, and i, and similarly for the modular flow, hence the image again lies in $\bigcap_\alpha \mathcal{M}_\alpha$. Minimality is immediate from the definition as the intersection of all subalgebras with the required properties. □

3.4 Main theorem

Theorem 3.8 (Two-source shared admissible manifest sector)**.** *Let $S^{(2)}_{\mathrm{multi}}$ be a two-source datum. Assume:*

1. ***Individual source closure.*** *Each source separately satisfies the named Foundation I hypotheses needed to produce*

$$(\mathcal{A}^{(1)}_c, \omega_1), \qquad (\mathcal{A}^{(2)}_c, \omega_2),$$

   together with foreground frames $\mathfrak{F}^{(1)}$ and $\mathfrak{F}^{(2)}$.

2. ***Joint-stability package.*** *There exists a candidate shared sector*

$$\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}} \subseteq \mathcal{A}^{(1)}_c \cap \mathcal{A}^{(2)}_c$$

   satisfying JS-A, JS-B, and JS-C.

3. ***Non-trivial shared foreground.*** *The shared foreground overlap $\mathfrak{F}_{\mathrm{sh}}$ is non-zero and admits a realization by exact shared pointer-aligned observables $O^{\mathrm{sh}}_1, \dots, O^{\mathrm{sh}}_{k_{\mathrm{sh}}}$ lying in $\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}}$.*

4. ***Admissibility input for Paper I.*** *The pair*

$$\left(\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}|_{\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})}\right)$$

   satisfies the admissibility conditions required by Paper I [2].

Then:

1. *the shared stable hull is well-defined and satisfies*

$$\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}} \subseteq \mathcal{A}^{(1)}_c \cap \mathcal{A}^{(2)}_c;$$

2. *the restricted state*

$$\omega^{\mathrm{st}}_{\mathrm{sh}} := \omega_{\mathrm{sh}}|_{\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})}$$

   is faithful on $\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})$;

3. *the pair*

$$\left(\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}), \omega^{\mathrm{st}}_{\mathrm{sh}}\right)$$

   is a shared admissible manifest sector;

4. therefore the Paper I inseparability theorem applies to that pair.

Proof. By assumption (3), the exact shared pointer-aligned observables $O_1^{\mathrm{sh}}, \dots, O_{k_{\mathrm{sh}}}^{\mathrm{sh}}$ lie in $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. Since $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$ is a von Neumann subalgebra by Definition 2.4, it contains the generated shared foreground algebra:

$$\mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}} \subseteq \mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}.$$

By assumption (2), the candidate shared sector satisfies JS-A, hence it is invariant under the ambient modular flow and under both source semigroups. Therefore

$$\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}} \in \mathfrak{L}_{\mathrm{sh}}^{\mathrm{st}}.$$

In particular, the class $\mathfrak{L}_{\mathrm{sh}}^{\mathrm{st}}$ is nonempty. Hence Definition 3.6 and Proposition 3.7 apply, so the shared stable hull is a well-defined von Neumann subalgebra of $\mathcal{A}_U$.

Because $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$ is a member of the shared stable-carrier class, minimality gives

$$\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}.$$

By assumption (2), the candidate shared sector is already contained in $\mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}$, so

$$\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

This proves part (1).

By JS-B, ω_{sh} is faithful on $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. The restriction of a faithful normal state to a von Neumann subalgebra is again faithful. Therefore

$$\omega_{\mathrm{sh}}^{\mathrm{st}} = \omega_{\mathrm{sh}}|_{\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}})}$$

is faithful on the shared stable hull. This proves part (2).

Parts (1) and (2), together with the stability properties from Proposition 3.7, show that the pair

$$\left(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}}\right)$$

is a shared manifest sector in the sense intended by the present paper. Assumption (4) is the external admissibility input that upgrades it to a shared *admissible* manifest sector, so Paper I applies. This proves parts (3) and (4). □

Corollary 3.9 (Shared inseparability)**.** *Under the hypotheses of Theorem 3.8, geometry and modular flow on the shared sector*

$$\left(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}}\right)$$

are not independently deformable in the sense of Paper I. In particular, any geometric reading $\mathrm{Geom}(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}})$ *defined within the Paper I theorem domain is a* shared *geometric sector.*

Proof. This is exactly the conclusion of the Paper I theorem applied to the pair isolated by Theorem 3.8. □

Remark 3.10 (What the main theorem does *not* prove)*.* Theorem 3.8 does not prove a general shared-world theorem. It proves something narrower: if a two-source datum already admits a candidate shared sector satisfying the named compatibility and admissibility inputs, then the overlap of the two source chains carries an honest theorem-level manifest sector to which the Paper I machinery may be applied. It also does not derive the candidate shared sector from multi-source entanglement or source dynamics alone. This is a foothold, not the full Bridge 2 closure.

Remark 3.11 (Where the theorem's new content lies). The genuine new content of Theorem 3.8 is the conditional inclusion chain

$$\text{shared foreground inclusion} \;+\; \text{JS-A} \Longrightarrow \mathcal{A}_{\text{sh}}^{\text{cand}} \in \mathfrak{L}_{\text{sh}}^{\text{st}} \Longrightarrow \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

The Paper I admissibility package is not derived here. It is a named external input specifying when the isolated shared stable hull lies in the theorem domain of Paper I.

4 Common/Private Decomposition

The main theorem isolates a shared admissible manifest sector. We now ask how each observer's full manifest algebra relates to that shared sector. In the Type III_1 setting, one should not treat this as a canonical factorization problem. The correct formulation is regime-dependent and uses a conditional expectation language.

Definition 4.1 (Shared/private regime assumption)**.** Let

$$\mathcal{A}_{\text{sh}} \subseteq \mathcal{A}_c^{(i)} \subseteq \mathcal{A}_U, \qquad i = 1, 2.$$

We say that the pair $(\mathcal{A}_c^{(i)}, \mathcal{A}_{\text{sh}})$ satisfies the *shared/private regime assumption* if for each i there exist:

1. a von Neumann subalgebra $\mathcal{A}_{\text{priv}}^{(i)} \subseteq \mathcal{A}_c^{(i)}$;
2. a faithful normal conditional expectation

$$E_{\text{sh}}^{(i)} : \mathcal{A}_c^{(i)} \to \mathcal{A}_{\text{sh}}$$

   preserving the individual manifest state in the sense that

$$\omega_i \circ E_{\text{sh}}^{(i)} = \omega_i;$$

3. a generating relation

$$\mathcal{A}_c^{(i)} = W^*\big(\mathcal{A}_{\text{sh}} \cup \mathcal{A}_{\text{priv}}^{(i)}\big).$$

Remark 4.2. Definition 4.1 does not assert uniqueness of the private complement, and it does not assert a tensor-product or free-product factorization. It only says that, under a regime assumption, the full manifest algebra can be generated from the shared sector together with a private complement, with a faithful conditional expectation projecting back to the shared sector.

Remark 4.3 (Minimal package for common/private language). Definition 4.1 is the minimal package needed before common/private language can even be used honestly in the present paper. Without a private complement, a faithful conditional expectation onto the shared sector, and a generating relation for the full manifest algebra, the phrase "private part" remains interpretive rather than mathematical.

Proposition 4.4 (Unpacking JS-B and localizing overlap discrepancy)**.** *Assume the hypotheses of Theorem 3.8, and suppose the shared/private regime assumption holds with*

$$\mathcal{A}_{\text{sh}} := \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}).$$

Let

$$\mathcal{A}_{\text{ov}} := \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

Define the overlap discrepancy functional

$$\delta_{12} := \omega_1|_{\mathcal{A}_{\text{ov}}} - \omega_2|_{\mathcal{A}_{\text{ov}}} \in (\mathcal{A}_{\text{ov}})_*.$$

Then:

1. *the two restricted manifest states agree on the shared sector:*

$$\omega_1|_{\mathcal{A}_{\text{sh}}} = \omega_2|_{\mathcal{A}_{\text{sh}}} = \omega_{\text{sh}}^{\text{st}}|_{\mathcal{A}_{\text{sh}}};$$

2. *equivalently,*

$$\delta_{12}|_{\mathcal{A}_{\text{sh}}} = 0;$$

3. *if $A \in \mathcal{A}_{\text{ov}}$ satisfies $\delta_{12}(A) \neq 0$, then $A \notin \mathcal{A}_{\text{sh}}$;*

Proof. For $A \in \mathcal{A}_{\text{sh}}$, JS-B clause (2), restricted from the candidate shared sector to the shared stable hull, gives

$$\omega_1(A) = \omega_{\text{sh}}^{\text{st}}(A) = \omega_2(A).$$

Therefore

$$\delta_{12}(A) = 0 \qquad \forall A \in \mathcal{A}_{\text{sh}}.$$

This proves the first two claims. The third is only the contrapositive of the second; it is not a derived theorem that independent states must agree on the whole overlap. □

Remark 4.5 (Logical status of Proposition 4.4). The agreement statement is an unpacking of the explicit JS-B assumption. The proposition's useful content is bookkeeping: once independent states are part of the datum, it identifies where a nonzero overlap discrepancy is compatible with that assumption. Proposition 2.17 shows that this nonzero case is not vacuous.

Corollary 4.6 (Shared manifest content)**.** *Under the hypotheses of Proposition 4.4, every observable in the shared sector carries common manifest content. Observer-dependent discrepancy, insofar as it is jointly accessible on $\mathcal{A}_{\text{ov}}$, can only appear outside the shared sector.*

Proof. Proposition 4.4 shows that the two restricted manifest states agree on the shared sector and that any jointly accessible disagreement must be detected off that sector. That is the formal meaning of "shared manifest content" used in the present paper. □

Remark 4.7 (Shared/private reading of Proposition 4.4). Under the shared/private regime assumption, Proposition 4.4 may be read as saying that any overlap-level disagreement is detected only outside the common manifest sector, i.e. on the observer-dependent side of the decomposition language introduced in this section.

Remark 4.8 (What Section 4 does *not* prove). Section 4 does not prove existence of a canonical private factor, nor does it derive the shared/private regime assumption from the bare two-source datum. Its role is narrower: once a shared manifest sector has been isolated and a regime assumption is granted, it makes precise where observer agreement must hold and where observer-dependent discrepancy may still appear.

Remark 4.9 (No canonical factorization claim). Corollary 4.6 should not be read as a theorem that

$$\mathcal{A}_c^{(i)} \cong \mathcal{A}_{\text{sh}} \bar{\otimes} \mathcal{A}_{\text{priv}}^{(i)}$$

or any analogous canonical factorization holds. The paper makes no such claim. It uses only the weaker and safer statement that, under a regime assumption, a shared sector and a private complement jointly generate the full manifest algebra.

5 Conditional N-Source Extension

The two-source theorem is the paper's main result. The natural next step is a conditional N-source extension. This section records the generalization in the same honest style as Foundation I: whatever can be internalized immediately is internalized, and the genuinely new input is kept explicit.

Definition 5.1 (Joint N-source Stage A compatibility)**.** Let

$$S_{\text{multi}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_i\}_{i=1}^N, \{\mathcal{E}_t^{(i)}\}_{i=1}^N)$$

be a multi-source datum with individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$. A family of Stage A projections $\{\mathcal{P}^{(i)}\}_{i=1}^N$ is jointly compatible with $\mathcal{A}_{\text{sh}}^{\text{cand}}$ if:

1. the normal idempotents $\mathcal{P}^{(i)}$ commute pairwise;
2. the joint product
$$\mathcal{P}_{\text{sh}}^{(N)} := \prod_{i=1}^N \mathcal{P}^{(i)}$$
has von Neumann-algebra range containing $\mathcal{A}_{\text{sh}}^{\text{cand}}$ and fixes that candidate sector pointwise;
3. the jointly projected predual dynamics admits one common kernel $\mathcal{K}_{\text{sh},*,N}(t,s)$ on the joint range.

Lemma 5.2 (Joint N-source Stage A object)**.** *Under Definition 5.1, $\mathcal{P}_{\text{sh}}^{(N)}$ is a normal idempotent and, together with the stipulated common kernel, defines a joint Stage A object.*

Proof. Pairwise commutation makes the finite product independent of order and gives

$$(\mathcal{P}_{\text{sh}}^{(N)})^2 = \prod_{i=1}^N (\mathcal{P}^{(i)})^2 = \mathcal{P}_{\text{sh}}^{(N)}.$$

Finite composition preserves normality. The range and common-kernel requirements are explicit clauses of Definition 5.1, so the resulting tuple has the required Stage A data. Pairwise two-source theorems alone would not supply this one joint kernel or guarantee that all pairwise overlap choices define the same range. □

Theorem 5.3 (Conditional N-source shared manifest sector)**.** *Let S_{multi} be an N-source datum with individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$. Assume:*

1. *there exists a candidate shared sector*
$$\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \bigcap_{i=1}^N \mathcal{A}_c^{(i)}$$
satisfying JS-A and JS-B for all sources;
2. *the joint Stage A compatibility condition of Definition 5.1 holds, and a compatible family of shared foreground overlaps generates one common foreground algebra $\mathcal{A}_{\mathfrak{F}_{\text{sh}}}$;*

3. *the N-source stable-carrier class*

$$\mathfrak{L}^{\mathrm{st}}_{\mathrm{sh},N} := \left\{ \mathcal{M} \subseteq \mathcal{A}_U : \begin{array}{l} \mathcal{M} \text{ is a von Neumann subalgebra}, \\ \mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}} \subseteq \mathcal{M}, \\ \sigma_s^{\omega_0,\mathrm{tot}}(\mathcal{M}) = \mathcal{M},\ \forall s, \\ \mathcal{E}_t^{(i)}(\mathcal{M}) \subseteq \mathcal{M},\ \forall t, \forall i \end{array} \right\}$$

is nonempty and contains $\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}}$;

4. *the resulting N-source shared stable hull satisfies the admissibility hypotheses needed by Paper I.*

Then the N-source shared stable hull is a shared admissible manifest sector contained in $\bigcap_{i=1}^{N} \mathcal{A}_c^{(i)}$.

Proof. Lemma 5.2 supplies the joint Stage A object that pairwise compatibility alone does not provide. The remaining proof is the same minimality argument as in the two-source case, with the only change that one now intersects over all source labels rather than over two. The candidate shared sector is, by assumption, a member of the N-source stable-carrier class. Therefore the intersection defining the N-source shared stable hull is well-defined and lies inside the candidate sector. The remaining steps repeat the proof of Theorem 3.8. □

Remark 5.4 (What Theorem 5.3 does not contain). Theorem 5.3 does not provide:

1. a uniqueness or maximality theorem for the N-source shared sector;
2. a quantitative relationship between the number/strength of sources and geometric stability;
3. a proof that the joint Stage A and stable-carrier conditions of Definition 5.1 follow from the bare multi-source datum.

All three belong to later work.

6 Open Problems

The conditional inputs of Foundation II are not cosmetic. They isolate the genuinely unfinished parts of the programme.

Heuristically, the present two-source theorem is closer to a mean-field foothold than to a full correlated treatment: it isolates a shared effective manifest sector in a restricted regime, while genuinely cross-source structure remains the central open problem for any future multi-source closure.

1. **Model-independent joint stability.** The present paper names JS-A, JS-B, and JS-C. It does not derive them model-independently from a bare multi-source datum. The analogue of Foundation I's observer-emergence bottleneck therefore persists in multi-source form.
2. **Compatible Nakajima–Zwanzig projections.** The present notion of compatible persistent projections is a new definition, not a standard theorem. A general existence theory for such compatible projections remains open.

3. **Shared-foreground non-triviality.** The paper proves that if a non-trivial shared foreground overlap exists, then its dimension is bounded by the individual foreground dimensions. Proposition 2.17 gives one FD/I example with nonzero overlap, but no Type III or model-independent theorem forces the overlap to be nonzero.

4. **Quantitative consensus versus stability.** The paper deliberately avoids a bound of the form

$$\text{consensus strength} \Longrightarrow \text{geometric fluctuation bound}.$$

That is the separate Bridge 3 problem.

5. N**-source non-conditional closure.** The present N-source theorem is conditional. A truly non-conditional closure would require deriving a joint N-fold Stage A projection, common kernel, and common foreground from source-side dynamics.

6. **Uniqueness and maximality of the shared sector.** Even in the two-source case, the present theorem proves only existence of a minimal shared stable hull inside a candidate shared carrier. It does not prove that this sector is unique among all plausible shared manifest sectors, or maximal in any global sense.

7. **Relation to Paper II.** Paper II constrains higher-order geometry by multi-directional consistency [3]. The present paper does not yet prove how the shared manifest sector of Foundation II interacts with the Paper II small-ball remainder structure. That interface remains open.

8. **Type III and infinite-dimensional obstacles.** The common/private regime assumption is not automatic in the Type III setting. Conditional expectations, complements, and generating decompositions all become more delicate in infinite dimensions. In particular, existence of a faithful normal conditional expectation onto a candidate shared sector typically interacts with modular invariance in the sense of Takesaki and may fail outside suitable invariant regimes. Even when such expectations exist, complements and generating relations need not be unique, and non-factorial or center-bearing sectors may force a central decomposition before any shared/private language becomes well behaved.

7 Conditional Input Ledger

This section records the named inputs that remain conditional in the present paper.

1. **I0: Independent source states.** The faithful normal states ϖ_i are independent datum, and $\omega_i = \varpi_i|_{\mathcal{A}_c^{(i)}}$. They are not automatic restrictions of the reference vector state ω_0.

2. **Independent-state rebuild after retraction.** The former single-ambient-state reading is withdrawn because it made discrepancy vanish on the entire overlap. The current conditional theorem uses independent ϖ_i, and shared agreement enters explicitly as JS-B rather than as a derived fact.

3. **I1: Individual source closure.** Each source separately satisfies the named Foundation I hypotheses needed to produce an individual manifest pair $(\mathcal{A}_c^{(i)}, \omega_i)$.

4. **JS-A: Shared-sector invariance and semigroup compatibility.** The candidate shared sector is stable under all relevant semigroups and under the ambient modular flow, and the restricted semigroups commute there.

5. **JS-B: Shared-state existence and faithfulness.** A common faithful normal state exists on the candidate shared sector and is compatible with the source-specific manifest states.

6. **JS-C: Compatible persistent projections.** The pair of source-side Nakajima–Zwanzig projections commute and admit a common projected kernel on the shared sector. Commuting conditional expectations supply its algebraic clauses, but not the kernel clause.

7. **Finite compatibility witness.** Proposition 2.17 verifies JS-A/B/C and $k_{\text{sh}} = 1$ in a finite dephasing model with independent states. It is a model check, not a Type III emergence theorem.

8. **F1: Non-trivial shared foreground with exact pointer realization.** The overlap of the two source-side foreground realizations is non-zero, finite dimensional, and admits an exact pointer-aligned realization inside the candidate shared sector.

9. **A1: Admissibility input for Paper I.** The shared stable hull satisfies the admissibility package needed for the geometry–modular-flow inseparability theorem.

10. **SP1: Shared/private regime assumption.** Each individual manifest algebra admits a faithful conditional expectation onto the shared sector together with a private complement that generates the full algebra jointly with the shared sector.

11. **N1: joint N-source Stage A condition.** One jointly commuting product projection, a common range, and one common projected kernel are required. Pairwise two-source compatibility is insufficient by itself. No unused global consensus functional is retained.

A Imported Interfaces

This appendix records, in theorem-ready form, the external interfaces imported from Genesis, Foundation I, and the Consciousness Trilogy. They are not reproved here, but they are restated explicitly so that the present paper can be read without reconstructing its dependencies from the earlier series papers. What is *not* imported should also be kept clear: the multi-source notions of candidate shared sector, JS-A, JS-B, JS-C, shared stable hull, and shared/private regime are defined internally in Foundation II itself.

Proposition A.1 (Foundation I single-source closure package)**.** *Foundation I [1] isolates a conditional chain*

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where:

1. *Stage A supplies a projected record object*

$$\mathcal{O}_A = (\mathcal{P}, \omega_t, \mathcal{K}_*(t,s)),$$

where $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$ is a normal idempotent projection and $\mathcal{K}_*(t,s)$ is a dual non-Markovian memory kernel;

2. *Stage B supplies a foreground frame $\mathfrak{F}$ with a finite-dimensional foreground space $V_{\text{fg}}(\mathfrak{F})$ and an operator realization*

$$\iota : V_{\text{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\text{sa}};$$

3. *Stage C supplies a manifest pair*

$$(\mathcal{A}_c, \omega) = \big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}), \omega\big),$$

where $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_U$ is a minimal modular-decoherence stable hull, hence a von Neumann subalgebra carrying a faithful normal restricted state;

4. *in the exact fixed-point comparison regime imported from Foundation I, one has the inclusion*

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\mathrm{HAFF}}.$$

(not used in the present paper; recorded for later reference).

Proposition A.2 (Imported single-source interfaces used in Foundation II)**.** *Whenever Foundation II invokes the i-th source chain, the only external single-source objects it uses are:*

1. *a Stage A projection $\mathcal{P}^{(i)}$, projected normal states $\omega_t^{(i)}$, and dual memory kernel $\mathcal{K}_{*,i}(t,s)$;*
2. *a Stage B foreground frame $\mathfrak{F}^{(i)}$ with realized finite-dimensional foreground image*

$$\iota_i\big(V_{\mathrm{fg}}(\mathfrak{F}^{(i)})\big) \subseteq \mathcal{A}_U^{\mathrm{sa}};$$

3. *a Stage C manifest pair*

$$(\mathcal{A}_c^{(i)}, \omega_i) = \big(\Phi^{\mathrm{st}}_{\mathfrak{F}^{(i)}}(\mathfrak{F}^{(i)}), \omega_i\big).$$

No multi-source compatibility theorem is imported at this stage. Foundation II introduces the multi-source compatibility layer itself.

Proposition A.3 (Paper I manifest inseparability theorem)**.** *Paper I [2] proves that for an admissible manifest pair $(\mathcal{A}_c, \omega)$, geometry and modular flow are not independently deformable. In particular, once a pair $(\mathcal{A}_c, \omega)$ in the theorem domain is given, the geometry and modular sectors should be treated as two constrained readings of one underlying algebraic datum. Foundation II uses this result only through its admissibility interface: whenever a shared stable hull is shown, by named external input, to belong to the Paper I theorem domain, the inseparability conclusion becomes available.*

Proposition A.4 (Paper II as a consistency constraint)**.** *Paper II [3] supplies a conditional CHM small-ball stress-tensor extraction. Its stronger Einstein-tensor identification requires additional two-derivative and transversality hypotheses and is not imported here. Foundation II uses the paper only as boundary calibration, not as a formal input to the shared-sector existence proof.*

Proposition A.5 (T-DOME source-side ingredients)**.** *The T-DOME papers motivate three single-source interfaces used upstream of the present paper:*

1. *T-DOME I motivates the Stage A projected record process with kernel $\mathcal{K}_*(t,s)$, but does not prove that non-Markovian memory is necessary for persistence [7, 10];*
2. *T-DOME II motivates finite-coordinate foreground compression. Rank saturation and symmetry breaking require the explicit model assumptions stated in Foundation I; they are not universal necessity theorems [8];*

3. *T-DOME III: self-calibration enters through a Fisher-information prior carried by the foreground structure as a specified model input, but no direct theorem from T-DOME III is used in the operator-algebraic proofs below beyond this source-side calibration role [9].*

Proposition A.6 (HAFF ambient setting)**.** *The HAFF programme supplies the ambient structural setting used here:*

1. *geometry is read from accessible algebraic structure rather than supplied as a background container [5];*

2. *accessible algebras are selected by stability under the relevant dynamics rather than by subjective choice [6];*

3. *in the exact fixed-point regime used for Foundation I and imported here, the HAFF accessible algebra*

$$\mathcal{A}_c^{\mathrm{HAFF}} \subseteq \mathcal{A}_U$$

   is a von Neumann subalgebra stable under the relevant semigroup and the ambient modular flow;

4. *fixed-point and stability ideas for completely positive semigroups motivate the use of stable carrier classes and exact pointer sectors in restricted regimes [16, 17].*

Remark A.7 (What earlier papers do and do not supply). The earlier Genesis and Consciousness papers supply only single-source objects and single-source comparison statements of the kind listed above. They do *not* supply a multi-source theorem asserting that two projections commute, that two foreground frames overlap nontrivially, or that two source chains admit a common stationary state. Those are the precise new conditional interfaces named JS-A, JS-B, and JS-C in the main text.

Remark A.8 (Conditional expectations). The common/private regime assumption uses faithful normal conditional expectations in the sense of Takesaki [13]. No claim is made here that such expectations always exist onto the shared sector in the Type III_1 setting; their existence is part of the regime assumption itself.

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Foundation III: The Perfuming Loop
A Conditional One-Step Source-Level Feedback Program

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Abstract

Foundation I [1] constructs a conditional single-source chain from a source datum S_{source} to an admissible manifest pair $(\mathcal{A}_c, \omega)$. Foundation II [2] extends this to two sources, isolating a shared manifest sector. Both chains are one-directional: source data produce manifest sectors, but the manifest sectors do not feed back to the source. The present paper, Foundation III, addresses the return arrow.

T-DOME III [6] supplies a calibration loop at the ego level under explicit residual-likelihood, convexity, drift, and hardware assumptions: the observer's prediction-error residuals drive a natural-gradient update of the reference-frame parameter σ, with tracking controlled by a local Lyapunov estimate. Paper III of the Consciousness Trilogy [5] now treats the former drift-floor theorem as withdrawn and leaves only a supplied arithmetic comparison score for finite self-models. Neither construction closes the loop at the source level: the ego adjusts its frame, but the decoherence semigroup $\{\mathcal{E}_t\}$ that generated the manifest sector in the first place is left untouched.

Foundation III defines a *source-level feedback map* that lifts the ego-level calibration signal to a controlled perturbation of the decoherence semigroup. Under named regularity and coupling conditions, we prove a conditional one-step closed-loop foothold with three components. First, the feedback-modified semigroup remains a valid source datum in the pointwise σ-weak Heisenberg-picture sense used by Foundation I v2. Second, the supplied self-description comparison functional under the closed loop satisfies a modified algebraic comparison bound that depends on the feedback coupling strength. Third, the one-directional Foundation I chain upgrades only to a conditional one-step closed loop:

$$\begin{aligned} S_{\text{source}} \longrightarrow \mathcal{O}_B \longrightarrow \text{calibration signal} \longrightarrow S_{\text{source}}[\text{updated}] \\ \longrightarrow \mathcal{O}_B[\text{updated}] \longrightarrow (\mathcal{A}_c, \omega)[\text{updated}]. \end{aligned}$$

The paper does not claim that the loop eliminates the self-description deficit. It proves only that the loop is well-posed and that the deficit transforms in a controlled way. The question of whether iterated perfuming converges to a fixed point, and what such a fixed point would mean structurally, is left as an open problem.

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1 Introduction

Foundation I [1] solved the first source-side bridge in the Consciousness Series. Starting from a source datum

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

it isolated a conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where $\mathcal{O}_A$ is a persistent projected record process, $\mathcal{O}_B$ is a foreground frame available under the explicit conditional coding interface of Foundation I, and $(\mathcal{A}_c, \omega)$ is the manifest accessible pair. The chain is one-directional: source data produce a manifest sector, and the series then hands that sector to Paper I of the Consciousness Trilogy [3] for the geometry–modular-flow theorem and to Paper II [4] for the higher-dimensional consistency analysis.

Foundation II [2] extended that source-side analysis to the first multi-source setting. There the main result was again one-directional. Given two source chains sharing one ambient Type III_1 substrate, one obtains a conditional inclusion theorem for a shared admissible manifest sector. What remained absent in both papers was any formal return arrow from manifest calibration back to the source dynamics that generated the manifest sector in the first place. Foundation II enters the present paper only as programme boundary calibration for the later multi-source feedback problem; no theorem below imports a Foundation II result.

The missing input now comes from two already published directions. T-DOME III [6] supplied a one-step calibration signal at the ego level: prediction-residual statistics determine a natural-gradient update $\delta\sigma$, controlled by a Lyapunov tracking estimate and penalized by the ego-rigidity term $\mathcal{I}_{\text{ego}}$. Paper III of the Consciousness Trilogy [5] proposed a local self-description floor on the manifest side. That reconstruction-loss bound has since been withdrawn: a lossless code for a low-dimensional orbit defeats an n-independent positive floor. Foundation III therefore uses its displayed Fisher expression only as a supplied comparison functional, not as an imported risk or reconstruction theorem. Neither that functional nor the T-DOME calibration model, by itself, modifies the source semigroup $\{\mathcal{E}_t\}$.

The present paper addresses exactly that gap. Its aim is not to prove a global convergence theorem for iterative self-modification or a general iterated-feedback theorem. It does not claim that the self-description deficit is eliminated by feedback, and it does not claim that one can derive a feedback lift from bare multi-scale dynamics without additional structure. Its narrower goal is to identify an honest one-step theorem-level foothold for *source-level feedback from manifest calibration*.

The main design choice is to avoid a formal expression of the type

$$\mathcal{E}_t \mapsto \mathcal{E}_t + \eta\,\Gamma(\delta\sigma, \mathcal{E}_t)$$

as the *definition* of the modified dynamics. Such a first-order expression is useful heuristically, but it does not by itself preserve the semigroup law or complete positivity. Instead, the paper defines a feedback lift through an exact auxiliary semigroup $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ of normal completely positive unital maps and then sets

$$\mathcal{E}_t^{[\eta]} := \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t.$$

Once a commutation condition is imposed, the semigroup law is automatic; the first-order feedback map Γ then reappears as the $\eta \downarrow 0$ tangent of this exact construction.

This paper therefore contributes four one-step theorem-level items.

1. We define a *source-level feedback lift* from an admissible calibration signal $\delta\sigma$ to an exact Heisenberg-picture semigroup perturbation on the ambient Type III_1 algebra.

2. We prove a preservation theorem: under named regularity and commutation inputs, the modified family $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ is again a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, so the ambient datum remains a valid source datum.

3. We prove a comparison theorem for a supplied local Fisher functional: if the chosen Foundation I chain is feedback-stable and its parameters remain controlled, then that functional deforms by an explicit small-coupling factor $f(\eta)$ with $f(0) = 1$. No actual deficit bound is inferred.

4. We package the result as a *one-step return datum*, showing that the chain

$$S_{\text{source}} \to \mathcal{O}_B \to \delta\sigma \to S_{\text{source}}^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$$

   is mathematically well-posed under named inputs.

The scope is deliberately narrow. The paper does not solve the constructive origin of the calibration signal, which is imported from T-DOME III. It does not solve the full persistence problem for the Foundation I chain under arbitrary perturbations, which remains a conditional interface. It does not derive explicit optimal coupling windows or fixed points for iterated perfuming. Those are precisely the open problems recorded later.

The paper also keeps the same Type III discipline as Foundation II. No tensor-product factorization is assumed. All dynamics are written in the Heisenberg picture on a common ambient algebra, and continuity means pointwise σ-weak continuity rather than norm continuity. This is not cosmetic: the whole point of the paper is to modify the source semigroup without importing an external finite-dimensional controller.

The paper is organized as follows. Section 2 defines the feedback lift, the regularity package, and the modified source datum. Section 3 proves the source-datum preservation theorem. Section 4 states the extra comparison inputs needed to pass through Foundation I and then proves the comparison-functional theorem. Section 5 packages the resulting one-step return map. Section 6 records the open problems. Section 7 collects the named conditional inputs. Appendix A restates the conditional Foundation I and T-DOME III interfaces and records the withdrawal of the former Paper III floor.

2 Feedback Lift and Regularity Conditions

The first task is to isolate the additional object that Foundation III needs and the exact conditions under which it is allowed to act on the source semigroup.

Definition 2.1 (Admissible calibration signal)**.** Fix a source datum S_{source} together with a chosen Foundation I chain up to Stage B, so that one has

$$\mathcal{O}_B = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s), \mathfrak{F}, \pi_{\text{ego}}).$$

An *admissible calibration signal* is a vector

$$\delta\sigma \in T_\sigma \Sigma$$

in the local calibration parameter space of T-DOME III such that:

1. $\delta\sigma$ is obtained from the natural-gradient calibration step imported from T-DOME III;
2. $\sigma + \delta\sigma$ remains inside the Lyapunov neighborhood on which the T-DOME III tracking estimate is valid;
3. there exists a bound $\|\delta\sigma\| \le R_{\text{cal}}$ for some finite calibration radius $R_{\text{cal}} > 0$.

Remark 2.2 (One-step rather than iterative calibration). Definition 2.1 is intentionally local. The paper treats a single feedback step driven by one calibration signal $\delta\sigma$. An iterated sequence $\delta\sigma_0, \delta\sigma_1, \ldots$ would require a separate fixed point or long-time theory, and nothing in the present paper should be read as supplying that theory.

Definition 2.3 (Feedback regularity package)**.** Let S_{source} be a source datum and let $\delta\sigma$ be an admissible calibration signal. A *feedback regularity package* consists of a family

$$\mathscr{F}^{[\delta\sigma]} = \{\mathscr{F}_r^{[\delta\sigma]}\}_{r\ge 0}$$

of normal maps on $\mathcal{A}_U$ satisfying the following conditions.

1. **FB-A: CP source lift.** $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\ge 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, with
$$\mathscr{F}_0^{[\delta\sigma]} = \mathrm{id}_{\mathcal{A}_U}, \qquad \mathscr{F}_r^{[0]} = \mathrm{id}_{\mathcal{A}_U} \quad \forall r \ge 0.$$
2. **FB-B: small-signal differentiability.** For every $A \in \mathcal{A}_U$, the map $r \mapsto \mathscr{F}_r^{[\delta\sigma]}(A)$ is σ-weakly differentiable at $r = 0$. We denote the derivative by
$$\mathfrak{G}_{\delta\sigma}(A) := \frac{d}{dr}\mathscr{F}_r^{[\delta\sigma]}(A)\Big|_{r=0},$$
and record the normalization $\mathfrak{G}_{\delta\sigma}(\mathbf{1}) = 0$, which follows from the unitality in FB-A because $\mathscr{F}_r^{[\delta\sigma]}(\mathbf{1}) = \mathbf{1}$ for all $r \ge 0$.
3. **FB-C: chain compatibility.** The feedback lift commutes with the ambient source semigroup:
$$\mathscr{F}_r^{[\delta\sigma]}\mathcal{E}_t = \mathcal{E}_t\mathscr{F}_r^{[\delta\sigma]} \qquad \forall r, t \ge 0.$$
In addition, for the chosen Foundation I chain, the maps $\mathscr{F}_r^{[\delta\sigma]}$ preserve the named structural inputs that are required later to reconstruct Stage A, Stage B, and the manifest pair.

Remark 2.4 (Why the feedback acts through an exact semigroup). The exact feedback semigroup in Definition 2.3 is not an ornament. A first-order perturbation $\mathcal{E}_t + \eta\,\Gamma(\delta\sigma, \mathcal{E}_t)$ does not automatically preserve the semigroup property or complete positivity. The exact semigroup lift is the cleanest way to keep those properties in the Heisenberg-picture Type III setting. The first-order map Γ reappears only after the exact object has been defined.

Remark 2.5 (Meaning of the strict commutation input). The commutation requirement in FB-C is strong and should be read carefully. It does *not* say that feedback is dynamically trivial. For $\eta > 0$,
$$\mathcal{E}_t^{[\eta]} = \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t$$
generally differs from $\mathcal{E}_t$ unless the calibration signal is zero or the coupling is turned off. The role of strict commutation is only to keep the algebra-level time ordering unchanged, so that the modified dynamics remains an exact semigroup without entering a more delicate generator-level perturbation problem. Relaxing FB-C to an approximate-commutation condition is left open.

Proposition 2.6 (A nontrivial commuting-dephasing feedback lift)**.** *The package FB-A–FB-C has a finite-dimensional normal UCP realization that depends nontrivially on $\delta\sigma$ and changes the source dynamics beyond a relabeling.*

Proof. Take $\mathcal{A}_U = M_2 \otimes M_2$, let Z_1, Z_2 be Pauli Z on the two factors, and write $\mathrm{Ad}_{Z_j}(A) = Z_j A Z_j$. Fix $\gamma, \kappa > 0$ and a continuous scalar response

$$q(\delta\sigma) := \|\delta\sigma\|^2.$$

Define bounded generators

$$\mathcal{L}_0 = \gamma(\mathrm{Ad}_{Z_1} - \mathrm{id}), \qquad \mathfrak{G}_{\delta\sigma} = \kappa q(\delta\sigma)(\mathrm{Ad}_{Z_2} - \mathrm{id}),$$

and semigroups

$$\mathcal{E}_t = e^{t\mathcal{L}_0}, \qquad \mathscr{F}_r^{[\delta\sigma]} = e^{r\mathfrak{G}_{\delta\sigma}}.$$

More explicitly,

$$\mathscr{F}_r^{[\delta\sigma]}(A) = a_r A + (1 - a_r) Z_2 A Z_2, \qquad a_r = \frac{1 + e^{-2\kappa q(\delta\sigma) r}}{2}.$$

Thus $\mathscr{F}_r^{[\delta\sigma]}$ is a normal UCP unital semigroup, equals the identity when $\delta\sigma = 0$, and has the stated generator at $r = 0$. The same formula with $(Z_2, \kappa q, r)$ replaced by (Z_1, γ, t) verifies these properties for $\mathcal{E}_t$.

Because Ad_{Z_1} and Ad_{Z_2} commute, so do the two semigroups, which is the algebraic clause of FB-C. A Foundation I input package supported on the joint diagonal algebra is fixed by both dephasings, so the named structural inputs can be chosen feedback-stable in this model. Yet for $A = \mathbf{1} \otimes X$,

$$\mathcal{E}_t(A) = A, \qquad \mathcal{E}_t^{[\eta]}(A) = e^{-2\eta\kappa q(\delta\sigma) t} A.$$

For $\eta > 0$ and $\delta\sigma \neq 0$, the lift therefore adds a real decay channel that the source semigroup did not have; it is not an inert renaming. This is an FD/I witness of compatibility, not a construction of a Type III feedback lift. □

Remark 2.7 (What strict commutation can and cannot add)*.* Proposition 2.6 shows that FB-C does not force the lift to be the identity. It also displays the limitation: the commuting generators are simultaneously diagonal on joint Pauli sectors, so feedback only reweights pre-existing joint modes and does not mix them. Creating new couplings between source sectors generally requires a near-commuting or time-ordered theory, which remains open.

Definition 2.8 (Small-coupling regime and modified semigroup)**.** Fix a feedback regularity package $\mathscr{F}^{[\delta\sigma]}$. A coupling strength $\eta \geq 0$ is in the *small-coupling regime* if $0 \leq \eta < \eta_*$ for some window $\eta_* > 0$ on which the later comparison inputs remain valid. For a first-order interpretation on a return horizon $[0, T]$, we additionally require $\eta T \ll 1$. The exact composed semigroup below exists without this perturbative estimate, but its tangent need not be uniformly small at long times. For such η, define the *feedback-modified semigroup* by

$$\mathcal{E}_t^{[\eta]} := \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t \qquad (t \geq 0).$$

The corresponding *modified source datum* is

$$S_{\text{source}}^{[\eta]} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t^{[\eta]}\}_{t \geq 0}).$$

Remark 2.9 (Heisenberg-picture convention). As in Foundation II, all semigroups in Definition 2.8 act on observables. Pointwise continuity therefore means σ-weak continuity on $\mathcal{A}_U$, not norm continuity. Nothing in the present paper uses or claims a norm-continuous generator-level perturbation theorem.

Definition 2.10 (First-order feedback map)**.** Under FB-B, define the *first-order feedback map* by

$$\Gamma(\delta\sigma, \mathcal{E}_t)(A) := t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)), \qquad A \in \mathcal{A}_U.$$

Equivalently, for each fixed $t \geq 0$,

$$\mathcal{E}_t^{[\eta]}(A) = \mathcal{E}_t(A) + \eta\, \Gamma(\delta\sigma, \mathcal{E}_t)(A) + o(\eta)$$

in the σ-weak topology as $\eta \downarrow 0$.

Verification of the expansion. By Definition 2.8,

$$\mathcal{E}_t^{[\eta]}(A) = \mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)).$$

Applying the $r = 0$ derivative from FB-B at $r = \eta t$ gives

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)) = \mathcal{E}_t(A) + \eta t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)) + o(\eta),$$

which is exactly the stated formula. □

Remark 2.11 (What Definition 2.10 does not prove). The first-order map Γ is merely the tangent of the exact feedback lift. Definition 2.10 does not claim that an arbitrary bounded normal map Γ integrates to a completely positive semigroup, nor does it characterize the most general generator-level perturbations compatible with Type III dynamics. Those questions are left open.

Remark 2.12 (The tangent is not uniform at long times). The identity

$$\Gamma(\delta\sigma, \mathcal{E}_t)(A) = t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A))$$

contains an explicit factor of t. Thus the first-order perturbation is controlled for fixed t as $\eta \downarrow 0$, or on windows with $\eta t \ll 1$ under a uniform generator bound. Small η alone does not give a perturbation uniformly small for arbitrarily long times. This is one reason the present result does not iterate to a long-time perfuming theorem.

3 Source-Datum Preservation

We now prove the first of the paper's three main results.

Lemma 3.1 (Semigroup law)**.** *Assume FB-A–FB-C. Then the family* $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ *from Definition 2.8 satisfies*

$$\mathcal{E}_t^{[\eta]}\mathcal{E}_s^{[\eta]} = \mathcal{E}_{t+s}^{[\eta]} \qquad \forall s, t \geq 0.$$

Proof. For $A \in \mathcal{A}_U$,

$$\begin{aligned}\mathcal{E}_t^{[\eta]}\mathcal{E}_s^{[\eta]}(A) &= \mathscr{F}_{\eta t}^{[\delta\sigma]}\, \mathcal{E}_t\Big(\mathscr{F}_{\eta s}^{[\delta\sigma]}\, \mathcal{E}_s(A)\Big) \\ &= \mathscr{F}_{\eta t}^{[\delta\sigma]}\mathscr{F}_{\eta s}^{[\delta\sigma]}\mathcal{E}_t\mathcal{E}_s(A),\end{aligned}$$

where FB-C is used to commute $\mathcal{E}_t$ past $\mathscr{F}_{\eta s}^{[\delta\sigma]}$. Since $\{\mathscr{F}_r^{[\delta\sigma]}\}$ and $\{\mathcal{E}_t\}$ are both semigroups,

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}\mathscr{F}_{\eta s}^{[\delta\sigma]}\mathcal{E}_t\mathcal{E}_s = \mathscr{F}_{\eta(t+s)}^{[\delta\sigma]}\mathcal{E}_{t+s} = \mathcal{E}_{t+s}^{[\eta]}.$$

□

Lemma 3.2 (Pointwise σ-weak continuity and positivity)**.** *Assume FB-A–FB-C. Then for every $A \in \mathcal{A}_U$,*

$$t \longmapsto \mathcal{E}_t^{[\eta]}(A)$$

is pointwise σ-weakly continuous. Moreover, each $\mathcal{E}_t^{[\eta]}$ is normal, completely positive, and unital.

Proof. Fix $A \in \mathcal{A}_U$ and a normal functional $\varphi \in (\mathcal{A}_U)_*$. By Definition 2.8,

$$\varphi(\mathcal{E}_t^{[\eta]}(A)) = \varphi\Big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\Big).$$

The map $t \mapsto \mathcal{E}_t(A)$ is σ-weakly continuous by the source-datum assumption, and $r \mapsto \mathscr{F}_r^{[\delta\sigma]}(B)$ is σ-weakly continuous for every fixed $B \in \mathcal{A}_U$ by FB-A. To prove continuity of the composition, fix $t_0 \geq 0$ and write

$$\begin{aligned}
&\Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t_0}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\quad\leq \Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\qquad+ \Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big) - \varphi\big(\mathscr{F}_{\eta t_0}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big|.
\end{aligned}$$

For the second term, $\mathcal{E}_{t_0}(A)$ is fixed, so pointwise σ-weak continuity of $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ gives convergence to zero as $t \to t_0$.

For the first term, use the predual semigroup $\{\mathscr{F}_{r,*}^{[\delta\sigma]}\}_{r\geq 0}$. Since $\{\mathscr{F}_r^{[\delta\sigma]}\}$ is a pointwise σ-weakly continuous semigroup of normal maps, the predual family is norm-continuous on $(\mathcal{A}_U)_*$. Thus

$$\begin{aligned}
&\Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\quad= \Big|\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big| \\
&\quad\leq \Big|\big(\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi) - \mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)\big)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big| \\
&\qquad+ \Big|\mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big|.
\end{aligned}$$

The first summand tends to zero because $\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi) \to \mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)$ in predual norm, while $\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)$ is norm-bounded by $2\|A\|$. The second summand tends to zero because $\mathcal{E}_t(A) \to \mathcal{E}_{t_0}(A)$ σ-weakly and $\mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)$ is a fixed normal functional. Hence the first term also tends to zero.

Hence $\{\mathcal{E}_t^{[\eta]}\}$ is pointwise σ-weakly continuous.

Normality and complete positivity are preserved under composition of normal completely positive maps, and unitality is preserved under composition of unital maps. Since both $\mathscr{F}_{\eta t}^{[\delta\sigma]}$ and $\mathcal{E}_t$ have those properties, so does $\mathcal{E}_t^{[\eta]}$. □

Theorem 3.3 (Source-datum preservation)**.** *Let*

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$$

be a source datum in the sense imported from Foundation I, let $\delta\sigma$ be an admissible calibration signal, and assume that a feedback regularity package FB-A–FB-C is given. Then for every $\eta \in [0, \eta_*)$, the modified datum

$$S_{\text{source}}^{[\eta]} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t^{[\eta]}\}_{t\geq 0})$$

is again a valid source datum: the family $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, and Ω remains cyclic and separating for $\mathcal{A}_U$.

Proof. Lemma 3.1 gives the semigroup law and Lemma 3.2 gives pointwise σ-weak continuity, normality, complete positivity, and unitality.

The ambient standard-form triple $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is not changed by the feedback step. In particular, Ω remains cyclic and separating for the same von Neumann algebra $\mathcal{A}_U$. Therefore $S_{\text{source}}^{[\eta]}$ satisfies the same source-datum format as S_{source}. □

Remark 3.4 (What Theorem 3.3 does not prove). The theorem proves only that the feedback-modified dynamics remains in the theorem domain of the Foundation I source-datum definition. It does not prove that the modified datum automatically reproduces the same persistent projection, the same foreground frame, or the same manifest algebra. Those are downstream interface questions and require additional inputs.

Corollary 3.5 (Exact CP lift with first-order tangent)**.** *Under the hypotheses of Theorem 3.3, the map $\Gamma(\delta\sigma, \mathcal{E}_t)$ from Definition 2.10 is the $\eta \downarrow 0$ tangent of an exact Heisenberg-picture completely positive perturbation.*

Proof. This is immediate from Theorem 3.3 together with the first-order expansion of Definition 2.10. □

4 Modified Local Comparison Functional

The preservation theorem keeps the modified dynamics inside the source-datum domain, but it does not yet produce a manifest pair. For that we need the chosen Foundation I chain to survive the perturbation and the supplied local comparison parameters to remain controlled.

Definition 4.1 (Deficit-comparison package)**.** Fix a source datum S_{source}, a chosen Foundation I chain from S_{source} to $(\mathcal{A}_c, \omega)$, and a feedback-modified datum $S_{\text{source}}^{[\eta]}$. We say that *DF-A* holds on the coupling window $[0, \eta_*)$ if:

1. **DF-A(1): feedback-stable Foundation I chain.** For every $\eta \in [0, \eta_*)$, the named conditional inputs used in the chosen Foundation I chain remain available for $S_{\text{source}}^{[\eta]}$. Hence one obtains a modified manifest pair
$$(\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

2. **DF-A(2): Fisher control.** There exist continuous functions $\alpha_F, \beta_{\text{ego}} : [0, \eta_*) \to (0, \infty)$ with
$$\alpha_F(0) = \beta_{\text{ego}}(0) = 1$$
such that
$$\mathcal{I}_F^{[\eta]} \geq \alpha_F(\eta)\, \mathcal{I}_F, \qquad \mathcal{I}_{\text{ego}}^{[\eta]} \leq \beta_{\text{ego}}(\eta)\, \mathcal{I}_{\text{ego}},$$
where $\mathcal{I}_F$ and $\mathcal{I}_{\text{ego}}$ are the baseline manifest-side Fisher sensitivity and ego-rigidity constants, and $\mathcal{I}_F^{[\eta]}$, $\mathcal{I}_{\text{ego}}^{[\eta]}$ are the corresponding modified quantities. Assume also
$$\mathcal{I}_F + \mathcal{I}_{\text{ego}} > 0, \qquad \mathcal{I}_F^{[\eta]} + \mathcal{I}_{\text{ego}}^{[\eta]} > 0,$$
so the comparison functionals are defined.

3. **DF-A(3): supplied local comparison parameters.** A controlled neighborhood is equipped with a stipulated parameter $\rho(\eta) \in [0, \frac{1}{2})$, continuous in η, with baseline value $\rho(0) =: \rho_0$. This parameter and the Fisher quantities define the comparison functional below; they do not import a valid Paper III reconstruction-loss theorem.

Proposition 4.2 (Passage from the modified source datum to the modified manifest pair)**.** *Assume the hypotheses of Theorem 3.3 and DF-A(1). Then for every $\eta \in [0, \eta_*)$, the modified source datum $S_{\text{source}}^{[\eta]}$ admits a conditional Foundation I chain*

$$S_{\text{source}}^{[\eta]} \to \mathcal{O}_A^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

Proof. Theorem 3.3 puts $S_{\text{source}}^{[\eta]}$ inside the theorem domain of the Foundation I source-datum definition. DF-A(1) states exactly that the named conditional inputs used by the chosen Foundation I chain remain available after the feedback perturbation. The Foundation I conditional theorem can therefore be re-run with the modified datum, yielding the stated outputs. □

Definition 4.3 (Baseline and modified local comparison functionals)**.** Under DF-A, define

$$\mathfrak{b}_{\text{loc}} := \frac{(1 - 2\rho_0)\, \mathcal{I}_F}{2\big(\mathcal{I}_F + \mathcal{I}_{\text{ego}}\big)},$$

and

$$\mathfrak{b}_{\text{loc}}^{[\eta]} := \frac{(1 - 2\rho(\eta))\, \mathcal{I}_F^{[\eta]}}{2\big(\mathcal{I}_F^{[\eta]} + \mathcal{I}_{\text{ego}}^{[\eta]}\big)}.$$

These are algebraic comparison scores. The formula is retained from the former Paper III interface, but neither score is asserted here to lower bound reconstruction loss, estimation risk, or a minimax deficit.

Theorem 4.4 (Comparison of supplied local functionals)**.** *Assume the hypotheses of Proposition 4.2 and the full package DF-A(1)–DF-A(3). Then for every $\eta \in [0, \eta_*)$,*

$$\mathfrak{b}_{\text{loc}}^{[\eta]} \geq f(\eta)\, \mathfrak{b}_{\text{loc}},$$

where

$$f(\eta) := \frac{1 - 2\rho(\eta)}{1 - 2\rho_0} \, \frac{\alpha_F(\eta)\big(\mathcal{I}_F + \mathcal{I}_{\text{ego}}\big)}{\alpha_F(\eta)\mathcal{I}_F + \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}}.$$

In particular, $f(0) = 1$, and by continuity $f(\eta) > 0$ for all sufficiently small η.

Proof. By Proposition 4.2, the modified datum yields a manifest pair $(\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$. For fixed $c > 0$ and $d \geq 0$, the function $x \mapsto cx/(x + d)$ is increasing where the denominator is nonzero. Therefore substituting the lower bound for $\mathcal{I}_F^{[\eta]}$ lowers the whole fraction, rather than creating a denominator sign reversal. Likewise the fraction is decreasing in its nonnegative second argument. Using DF-A(2),

$$\mathcal{I}_F^{[\eta]} \geq \alpha_F(\eta)\mathcal{I}_F, \qquad \mathcal{I}_{\text{ego}}^{[\eta]} \leq \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}},$$

so

$$\mathfrak{b}_{\text{loc}}^{[\eta]} \geq \frac{(1 - 2\rho(\eta))\, \alpha_F(\eta)\mathcal{I}_F}{2\big(\alpha_F(\eta)\mathcal{I}_F + \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}\big)}.$$

Now factor out the baseline score from Definition 4.3:

$$\begin{aligned}\mathfrak{b}_{\text{loc}}^{[\eta]} &\geq \frac{1-2\rho(\eta)}{1-2\rho_0}\,\frac{\alpha_F(\eta)\big(\mathcal{I}_F+\mathcal{I}_{\text{ego}}\big)}{\alpha_F(\eta)\mathcal{I}_F+\beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}}\cdot\frac{(1-2\rho_0)\mathcal{I}_F}{2(\mathcal{I}_F+\mathcal{I}_{\text{ego}})}\\ &= f(\eta)\,\mathfrak{b}_{\text{loc}}.\end{aligned}$$

The formula for $f(0)$ follows from $\rho(0)=\rho_0$ and $\alpha_F(0)=\beta_{\text{ego}}(0)=1$. Continuity then gives positivity for sufficiently small η. □

Remark 4.5 (No monotonicity claim for $f(\eta)$). The theorem asserts only that $f(0)=1$ and that $f(\eta)$ remains positive for sufficiently small coupling. No monotonicity statement is claimed. Depending on the detailed response of the manifest Fisher sensitivity and the ego-rigidity term, one may have $f(\eta)<1$ or $f(\eta)>1$. In particular, a feedback step that increases $\mathcal{I}_F^{[\eta]}$ or decreases $\mathcal{I}_{\text{ego}}^{[\eta]}$ can raise the score rather than lower it.

Remark 4.6 (The former actual-deficit corollary is withdrawn). No saturation assumption can repair the missing premise that $\mathfrak{b}_{\text{loc}}$ is a valid positive lower bound on the actual reconstruction loss. Establishing such a bridge requires either a specified sampling/measurement model and an estimation-risk theorem, or a genuine rate-distortion bottleneck. The present paper supplies neither, so it makes no inequality claim for an actual minimax deficit.

Remark 4.7 (What Theorem 4.4 does not prove). The theorem gives a controlled deformation of a supplied algebraic score. It does not prove that the true minimax deficit is monotone in η, it does not prove that feedback reduces the deficit, and it does not prove that the deficit can be driven to zero. Those stronger claims would require a valid risk/distortion model and a transport theory for optimal self-models.

Remark 4.8 (Why the comparison factor is the right theorem target). The factor $f(\eta)$ isolates exactly what feedback changes at theorem level:

1. the supplied local coefficient $\rho(\eta)$;
2. the manifest Fisher sensitivity $\mathcal{I}_F^{[\eta]}$;
3. the ego-rigidity contribution $\mathcal{I}_{\text{ego}}^{[\eta]}$.

No stronger conclusion is claimed. In particular, the theorem does not say that a better calibration signal necessarily increases $\mathcal{I}_F^{[\eta]}$ or decreases $\mathcal{I}_{\text{ego}}^{[\eta]}$; it only says that if those quantities remain controlled, then the comparison score remains controlled as well.

5 Conditionally Well-Posed One-Step Return Map

We can now state the paper's final theorem.

Definition 5.1 (One-step return datum)**.** Under the hypotheses of Theorem 4.4, the *one-step return datum at coupling* η is the tuple

$$\mathfrak{C}_\eta := \big(S_{\text{source}}, \mathcal{O}_B, (\mathcal{A}_c,\omega), \delta\sigma, S_{\text{source}}^{[\eta]}, \mathcal{O}_B^{[\eta]}, (\mathcal{A}_c^{[\eta]},\omega^{[\eta]})\big),$$

where $\mathcal{O}_B$ and $(\mathcal{A}_c,\omega)$ are the baseline Stage B and Stage C outputs of the chosen Foundation I chain, $\delta\sigma$ is the admissible calibration signal extracted from the retained Stage B data, and $\mathcal{O}_B^{[\eta]}$ together with $(\mathcal{A}_c^{[\eta]},\omega^{[\eta]})$ are the corresponding modified outputs after the feedback step.

Theorem 5.2 (Conditional one-step return-map well-posedness)**.** *Let S_{source} be a source datum, let $\delta\sigma$ be an admissible calibration signal, and assume the packages FB-A–FB-C and DF-A(1)–DF-A(3). Then for every sufficiently small $\eta \in [0, \eta_*)$, and on any fixed return horizon $[0, T]$ with $\eta T \ll 1$, the one-step return datum $\mathfrak{C}_\eta$ of Definition 5.1 is well-defined. Equivalently, the diagram*

$$S_{\mathrm{source}} \longrightarrow \mathcal{O}_B \longrightarrow \delta\sigma \longrightarrow S_{\mathrm{source}}^{[\eta]} \longrightarrow \mathcal{O}_B^{[\eta]} \longrightarrow (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$$

is a conditionally well-posed one-step return into the source-datum category.

Proof. The baseline arrow

$$S_{\mathrm{source}} \to \mathcal{O}_B \to (\mathcal{A}_c, \omega)$$

is the chosen Foundation I chain. The arrow

$$\mathcal{O}_B \to \delta\sigma$$

is the imported T-DOME III calibration interface. By Theorem 3.3, the feedback step

$$\delta\sigma \to S_{\mathrm{source}}^{[\eta]}$$

is well-defined at the source-datum level. By Proposition 4.2, the same conditional Foundation I chain can then be run again on $S_{\mathrm{source}}^{[\eta]}$, yielding

$$S_{\mathrm{source}}^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

Thus every component of $\mathfrak{C}_\eta$ is defined. The supplied comparison score is controlled by Theorem 4.4. This does not turn the one-step return into a self-consistent loop or an actual deficit theorem. □

Corollary 5.3 (Packaged one-step return map)**.** *Under the hypotheses of Theorem 5.2, there is a well-defined map*

$$\mathscr{L}_\eta : \big(S_{\mathrm{source}}, \delta\sigma\big) \longmapsto \big(S_{\mathrm{source}}^{[\eta]}, \mathcal{O}_B^{[\eta]}, (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}), \mathfrak{b}_{\mathrm{loc}}^{[\eta]}\big).$$

Proof. This is just a repackaging of Theorem 5.2 and Theorem 4.4. □

Remark 5.4 (What Theorem 5.2 does not prove). The theorem is a one-step well-posedness statement only. It does not claim that repeated application of $\mathscr{L}_\eta$ converges, that a fixed point exists, that a fixed point would be unique, or that the return map is thermodynamically optimal. It also does not address the multi-source feedback problem in which several manifest sectors update a shared ambient semigroup simultaneously.

6 Open Problems

The present note defines only a conditionally well-posed one-step return. The actual programme still leaves several substantive gaps.

1. **Constructive origin of a Type III feedback lift.** Proposition 2.6 supplies a nontrivial FD/I witness, but the general theorem still takes the source-level semigroup lift $\mathscr{F}^{[\delta\sigma]}$ as named input. A future result must derive a normal Type III lift from the calibration signal itself.

2. **Generator-level perturbation theory.** The present paper works with an exact auxiliary semigroup because it preserves complete positivity and the semigroup law cleanly. A sharper generator-level perturbation theorem on W^*-algebras, closer to the existing structural theory of dynamical semigroups on W^*-algebras [10], would have to control densely defined predual generators, domain stability, closability, and resolvent bounds in a regime where pointwise σ-weak continuity is the natural topology and norm continuity is usually unavailable. Naive first-order generator perturbations may fail to preserve complete positivity, unitality, or even semigroup well-posedness.

3. **Weakening the commutation input.** FB-C requires strict algebra-level commutation between the feedback lift and the source semigroup. A weaker theorem based on approximate commutation, asymptotic commutation, or commutation only on the relevant stable subalgebras would be more flexible and physically more natural, but it would need a replacement for the exact time-ordering argument used in Section 3. The finite witness shows explicitly that strict commutation can change decay rates while still only reweighting joint spectral sectors.

4. **Quantitative coupling thresholds.** The small-coupling window $[0, \eta_*)$ is left abstract. The programme still lacks an explicit estimate of η_* in terms of Lyapunov constants, semigroup norms, and the local BKM/Fisher geometry.

5. **Persistence of the Foundation I chain.** DF-A(1) is a real interface burden. One needs a theorem stating when persistent projections, foreground frames, and stable-hull inputs are robust under source-side feedback.

6. **A valid risk or distortion interface.** Theorem 4.4 controls only the supplied score $\mathfrak{b}_{\mathrm{loc}}$. A sampling model with an estimation-risk theorem, or a rate-distortion model with a genuine bottleneck, is required before any actual minimax deficit can be transported.

7. **Iterated perfuming and fixed points.** Once $\mathscr{L}_\eta$ is applied repeatedly, one faces an entirely new dynamical question: do the semigroups converge, cycle, or wander? The present paper does not address that long-time behavior.

8. **Thermodynamic cost of the source-level return.** T-DOME III gives model-dependent reset costs for the calibration procedure at the manifest/ego level. Foundation III has not integrated a specified irreversible implementation with the source-side semigroup perturbation.

9. **Multi-source feedback.** Foundation II handled the two-source shared-sector problem at inclusion level, but it did not address simultaneous feedback from multiple manifest sectors onto one common ambient source dynamics. Cross-source feedback remains open.

10. **Type III modular obstacles.** Even if a feedback lift is available, one still faces the usual Type III issues: modular invariance of the relevant subalgebras, possible non-uniqueness of conditional expectations, central-decomposition effects in non-factorial regimes, and the absence of a canonical bounded algebra-side generator picture that could be perturbed by finite-rank feedback data.

7 Conditional Input Ledger

The theorem chain depends on the following named inputs.

1. **FI-A: Foundation I source-to-manifest interface.** The single-source conditional chain from a source datum to a manifest pair is available in the sense restated in Appendix A.

2. **TD3-A: admissible calibration signal.** A one-step natural-gradient update signal $\delta\sigma$ exists and lies inside the T-DOME III Lyapunov neighborhood.

3. **FB-A: CP source lift.** The calibration signal admits a pointwise σ-weakly continuous semigroup lift by normal completely positive unital maps on $\mathcal{A}_U$.

4. **FB-B: small-signal differentiability.** The lifted feedback semigroup is differentiable at zero coupling so that the first-order map Γ is defined; its normalization $\mathfrak{G}_{\delta\sigma}(\mathbf{1}) = 0$ is inherited from unitality.

5. **FB-C: chain compatibility.** The feedback lift commutes with the ambient source semigroup and preserves the named structural inputs needed to rerun the chosen Foundation I chain.

6. **FB-FD: finite commuting-dephasing witness.** Proposition 2.6 verifies nontrivial $\delta\sigma$-dependence, normal UCP dynamics, strict commutation, and a source-dynamical change. It is a model check, not a Type III construction.

7. **DF-A(1): feedback-stable Foundation I chain.** The chosen Foundation I conditional inputs remain valid for $S^{[\eta]}_{\text{source}}$ on a nontrivial coupling window.

8. **DF-A(2): manifest Fisher control.** The modified manifest-side Fisher sensitivity and ego-rigidity terms satisfy the comparison inequalities controlled by $\alpha_F(\eta)$ and $\beta_{\text{ego}}(\eta)$.

9. **DF-A(3): supplied local comparison parameters.** The coefficient $\rho(\eta) < \frac{1}{2}$ and the Fisher quantities define $\mathfrak{b}^{[\eta]}_{\text{loc}}$; no Paper III reconstruction-loss theorem is inherited.

10. **Downgraded one-step return status.** The theorem-level output is a conditionally well-posed one-step return map. Its tangent is controlled only at fixed t as $\eta \downarrow 0$, or on windows $\eta t \ll 1$. Iteration, convergence, and fixed points remain open.

A Imported Interfaces

This appendix records the external results used by the body in a self-contained theorem-ready form. The statements are imported interfaces, not re-proved results.

Remark A.1 (What is and is not imported). Foundation III imports a conditional source-to-manifest interface from Foundation I and a model-level calibration interface from T-DOME III. It does not import the withdrawn Paper III reconstruction-loss theorem; only the algebraic comparison formula is retained as supplied data. Foundation II is cited in the Introduction solely to locate the later multi-source feedback problem and does not enter any proof or named input in the present paper.

Proposition A.2 (Imported Foundation I interface)**.** *Let*

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$$

be a source datum, where $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a Type III$_1$ standard-form von Neumann algebra and $\{\mathcal{E}_t\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$. Suppose:

1. *there exists a persistent Nakajima–Zwanzig projection $\mathcal{P}$ whose projected predual dynamics admits a memory kernel $\mathcal{K}_*(t,s)$ [8, 9, 7];*

2. *a specified finite-coordinate coding model satisfies the Foundation I rank-cost, strict-improvement, and group-action assumptions and supplies a compressed foreground frame $\mathfrak{F}$;*

3. *the foreground transfer/stable-hull construction closes so as to produce a von Neumann algebra $\mathcal{A}_c \subseteq \mathcal{A}_U$ with faithful restricted state $\omega = \langle \Omega, \cdot\, \Omega\rangle|_{\mathcal{A}_c}$;*

4. *the admissibility input needed to feed Paper I [3] is available.*

Then the conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega)$$

is defined.

Remark A.3. Foundation III uses only the existence of the manifest pair produced by Proposition A.2. It does not revisit the internal stable-hull construction of Foundation I. The source-datum continuity imported here is pointwise σ-weak continuity, matching Foundation I v2. The published Foundation I v1 uses the stronger phrase "strongly continuous"; the downstream Foundation I chain used here does not depend on that stronger condition.

Proposition A.4 (Conditional T-DOME III calibration interface)**.** *Let $\mathcal{O}_B$ be a Stage B output in the sense of Foundation I. In the local calibration regime of T-DOME III there exist:*

1. *a natural-gradient update signal $\delta\sigma \in T_\sigma\Sigma$;*

2. *a Lyapunov neighborhood $\mathcal{U}_{\text{cal}}$ on which the one-step update stays controlled;*

3. *a regular prior Fisher term $\mathcal{I}_{\text{ego}} \geq 0$, interpreted as ego rigidity, for the specified local estimation model.*

These data determine the one-step manifest-side calibration step used in the body of the present paper.

Remark A.5. Proposition A.4 is a one-step interface only. It does not assert global convergence of repeated calibration updates.

Definition A.6 (Supplied local Fisher comparison score)**.** For stipulated nonnegative quantities $\mathcal{I}_F$ and $\mathcal{I}_{\text{ego}}$, and $\rho \in [0, \frac{1}{2})$, define

$$\mathfrak{b}_{\text{loc}} := \frac{(1-2\rho)\,\mathcal{I}_F}{2(\mathcal{I}_F + \mathcal{I}_{\text{ego}})}.$$

The definition makes no assertion that this score lower-bounds an actual reconstruction loss or estimation risk.

Remark A.7 (Status of the former Paper III import). The original positive drift-floor theorem is withdrawn because its admissible reconstruction class includes lossless low-dimensional orbit codes. Definition A.6 retains only the algebraic expression needed for Theorem 4.4. A future sampling model may support an estimation-risk bound, and a future bottleneck may support a rate-distortion bound, but neither result is assumed here.

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